Leachate Analysis from 2025 Greenhouse Study

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GitHub repository

Methods

Sample Collection and Analysis

Nutrient leachate samples were collected at five timepoints throughout the 14-week wheat growing period: weeks 2, 4, 6, and 8 (corresponding to days 14, 28, 42, and 56 post-planting). To collect leachate, each pot was watered heavily with 1000 mL of water. After allowing drainage, leachate was collected from the collection tray beneath each pot. All seven biological replicates per treatment were sampled at each timepoint (n = 7 per treatment per timepoint).

Leachate samples were analyzed for ammonium (NH -N), nitrate (NO -N), and phosphate (PO^3-P) concentrations using a Gallery™ Analyzer (ThermoFisher Scientific). A water blank was included in each analytical run for quality control. Due to the sensitivity of phosphate to freezing, samples were stored at 4°C and analyzed within 24 hours of collection to prevent degradation.

Data Analysis

General trends in nutrient leachate data (NO , NH , PO^3) were visualized using violin plots.

Data were analyzed separately for each sampling week using one-way analysis of variance (ANOVA) with treatment as the fixed factor. Prior to analysis, ANOVA assumptions were verified by visual inspection of residual plots and Shapiro-Wilk tests for normality. For datasets meeting parametric assumptions, Tukey's Honest Significant Difference (HSD) post-hoc tests were conducted to identify pairwise differences among treatments (= 0.05).

All analyses were conducted in R version 2025.09.2+418 using the base stats package, tidyverse, dplyr. Plots were generated with the ggstatsplot, ggplot2, plotrix, and ggpubr.

Hypotheses

1. Treatments with nitrogen-fixing bacteria (T1, T3, T4, T5, T7, T9) will exhibit elevated ammonium leachate concentrations compared to untreated controls due to enhanced N fixation in the plant-soil system.
2. Treatments containing AMF (T2, T3, T4, T6) will show reduced phosphate leachate compared to non-AMF treatments due to direct P transfer to plants via mycorrhizal networks.

3. Combined bacteria+AMF treatments (T3, T4) will demonstrate synergistic effects, showing lower leachate concentrations of all three nutrients compared to single-inoculant treatments due to complementary nutrient acquisition strategies.

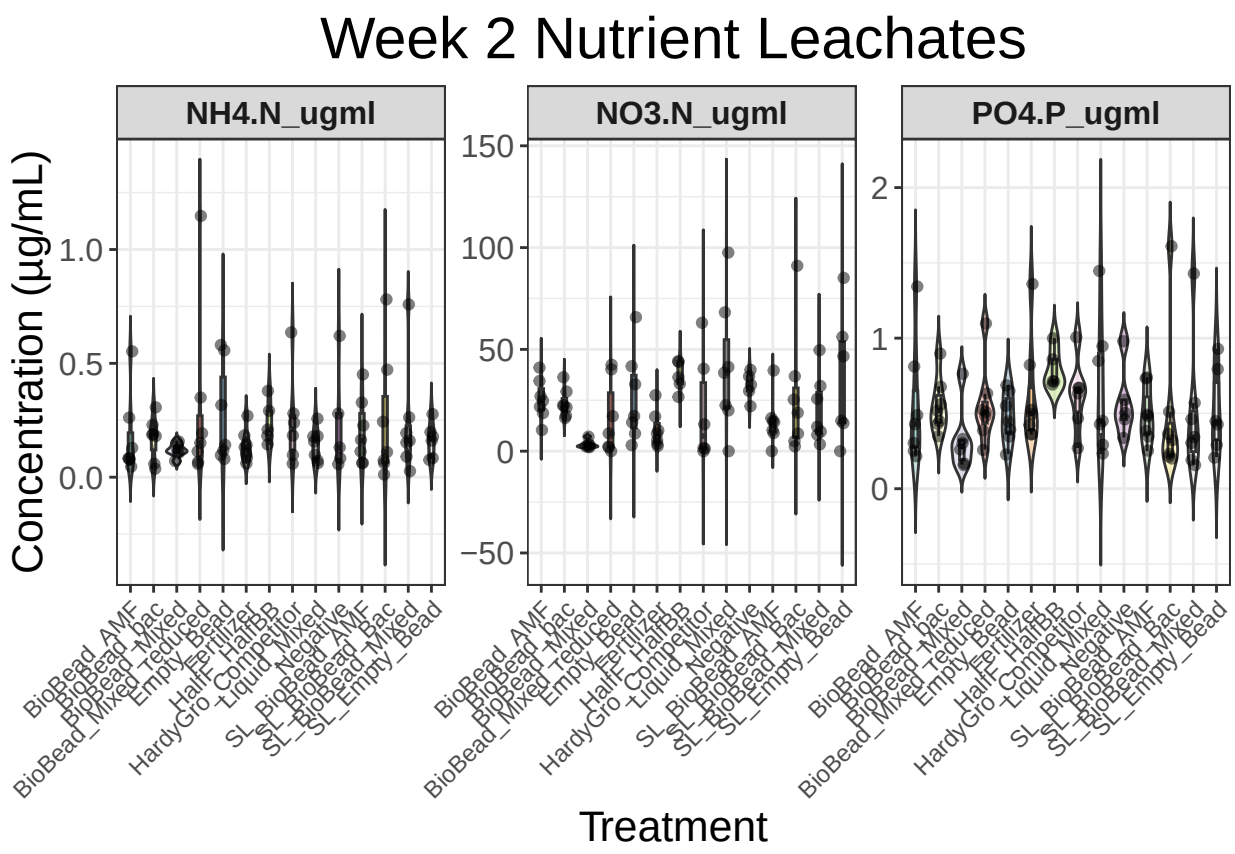
4. Strigolactone-primed treatments (T4, T5, T6) will exhibit lower nutrient leachate at later timepoints (weeks 6-8) compared to non-primed equivalents due to accelerated AMF colonization and symbiosis establishment.

5. Synthetic fertilizer (T13) will show highest nutrient leachate concentrations, particularly early in the study, due to rapid nutrient availability exceeding plant uptake capacity.

Results

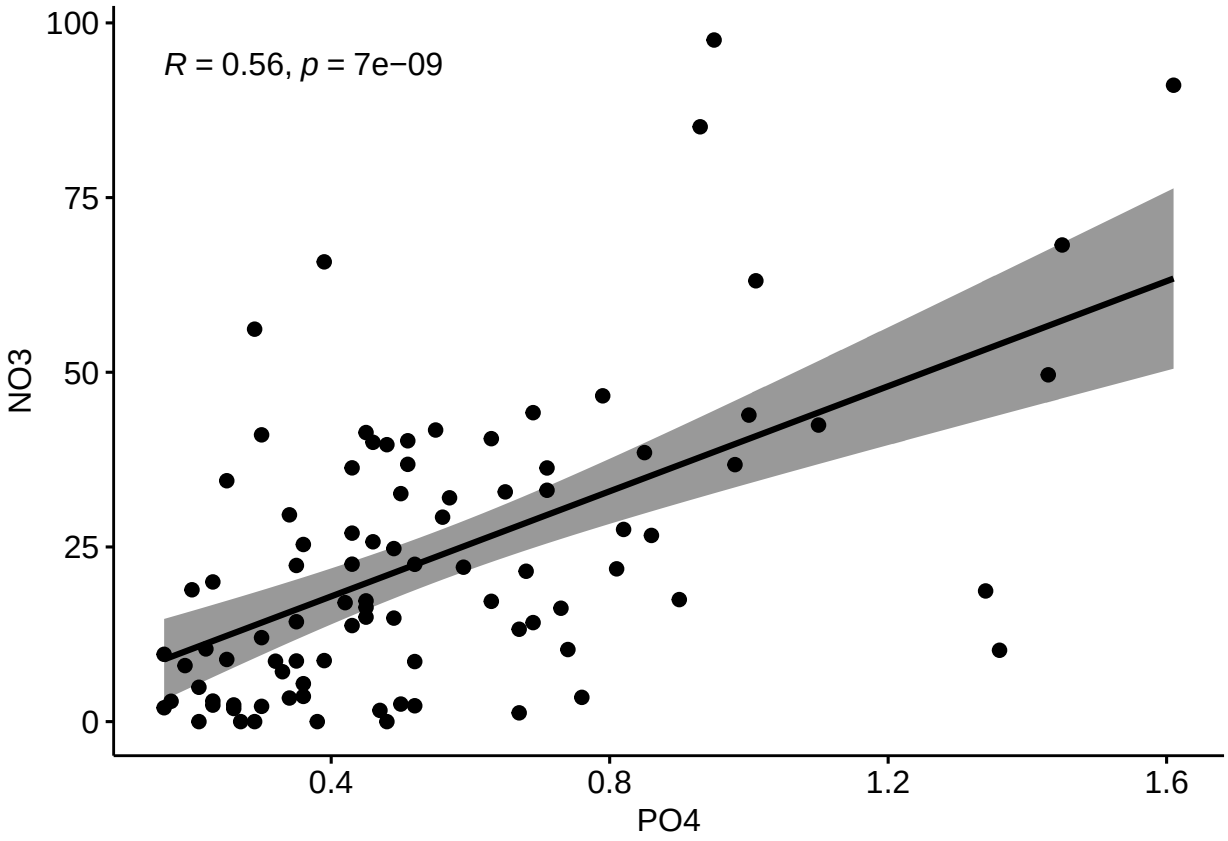
Week 2 Analysis

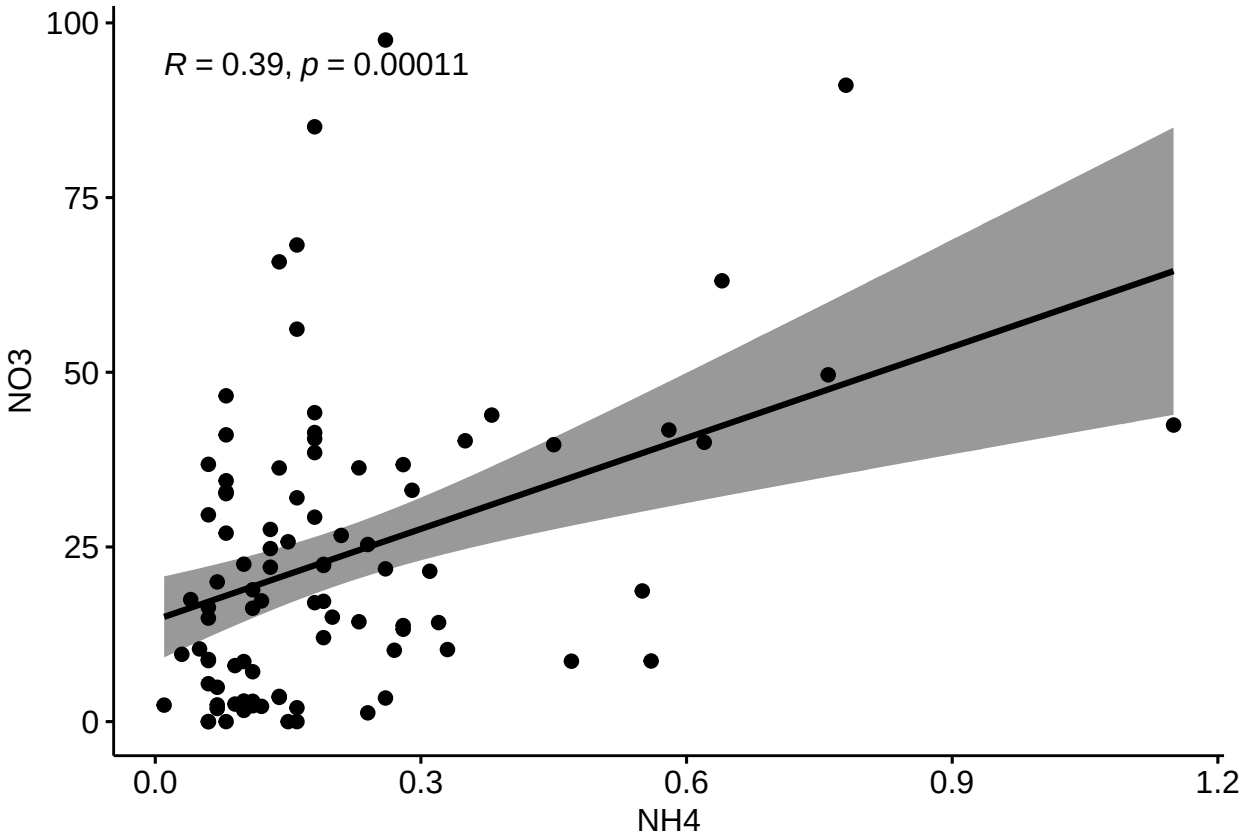
Faceted violin plot



Correlation scatterplots

Correlation scatterplots were generated to assess the correlations between nitrate and phosphate and ammonium.


R = 0.56, p = 7e−09
100
75
50
25
0
NO3
0.4
0.8
1.2
1.6
PO4



One Way ANOVA

```
summary(anova_NH4_wk2)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13 0.2583 0.01987   0.517  0.908
## Residuals          77 2.9601 0.03844
```

```
summary(anova_NO3_wk2)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13   9209   708.4   1.821 0.0544 .
## Residuals          77  29947   388.9
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(anova_PO4_wk2)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  0.868 0.06675   0.653  0.801
## Residuals          77  7.866 0.10216
```

There were no significant differences among treatment groups for ammonium and phosphate concentrations in leachate. Nitrate concentrations differed marginally between treatments (0.0544).

Tukey Test

There were no significant differences among treatment groups for ammonium and phosphate concentrations in leachate. Nitrate concentrations different marginally between Liquid_Mixed vs BioBead_Mixed ($p = 0.0523$), with Liquid_Mixed associated with higher concenrations.

Shapiro Wilk test for normality

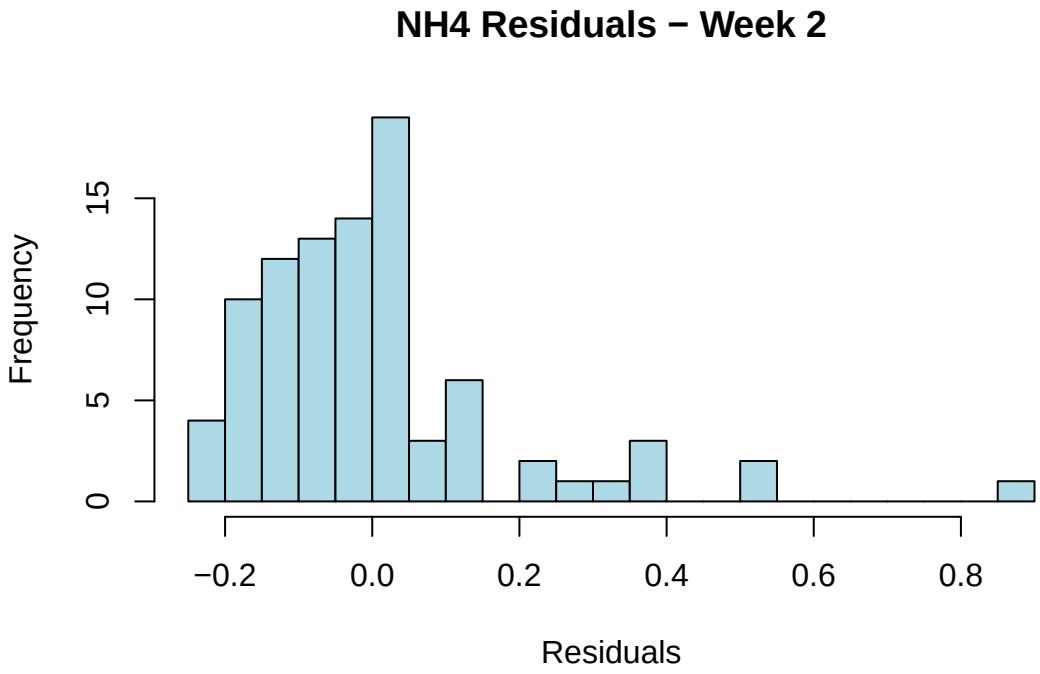
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NH4_wk2$residuals
## W = 0.83315, p-value = 9.807e-09
```

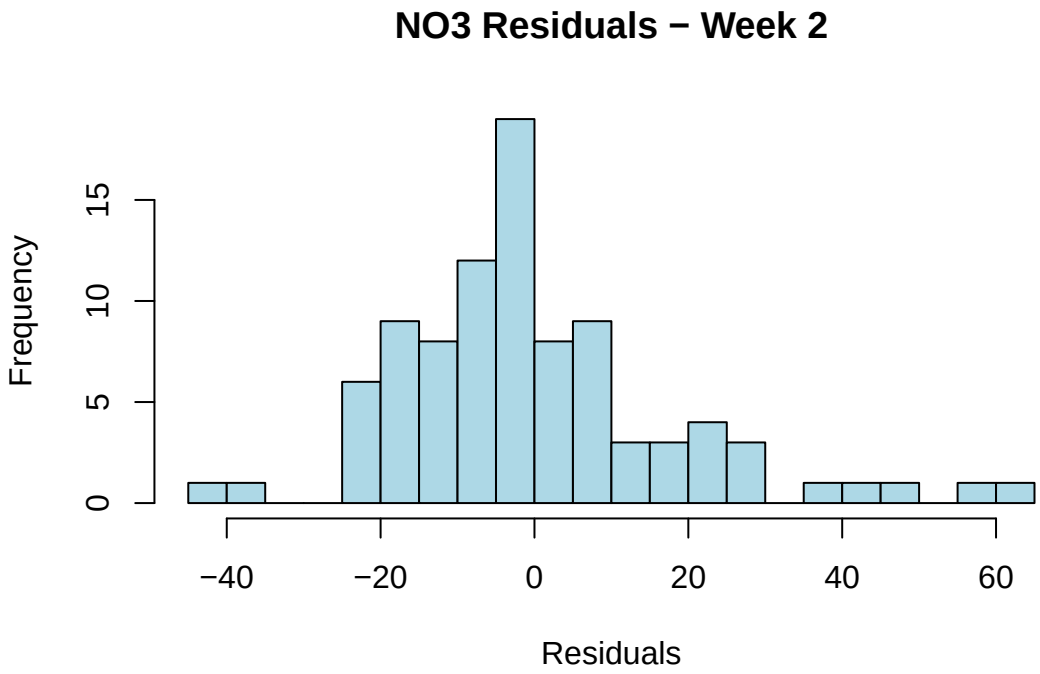
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NO3_wk2$residuals
## W = 0.92787, p-value = 8.344e-05
```

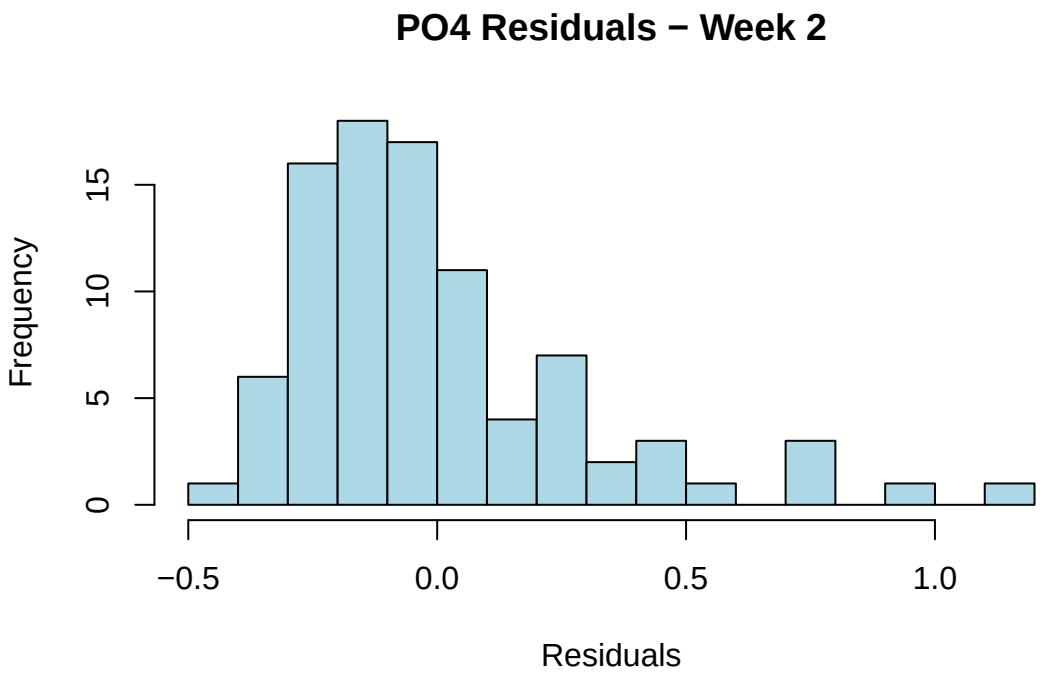
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_PO4_wk2$residuals
## W = 0.86564, p-value = 1.393e-07
```

Together, the W values for all ANOVA tests (0.83-0.93) indicate deviation from normality. The very low p values indicate that there are significant differences in normality. Therefore, alternative statistical analyses should be considered.

Residuals distribution

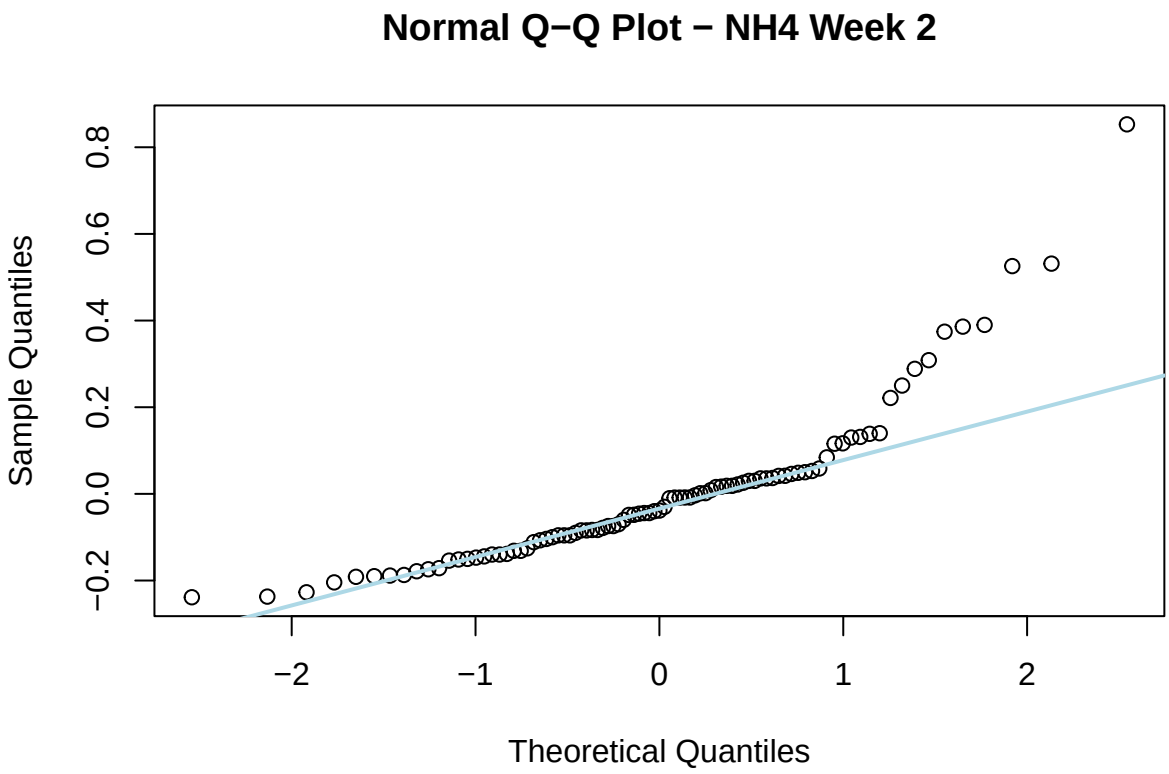




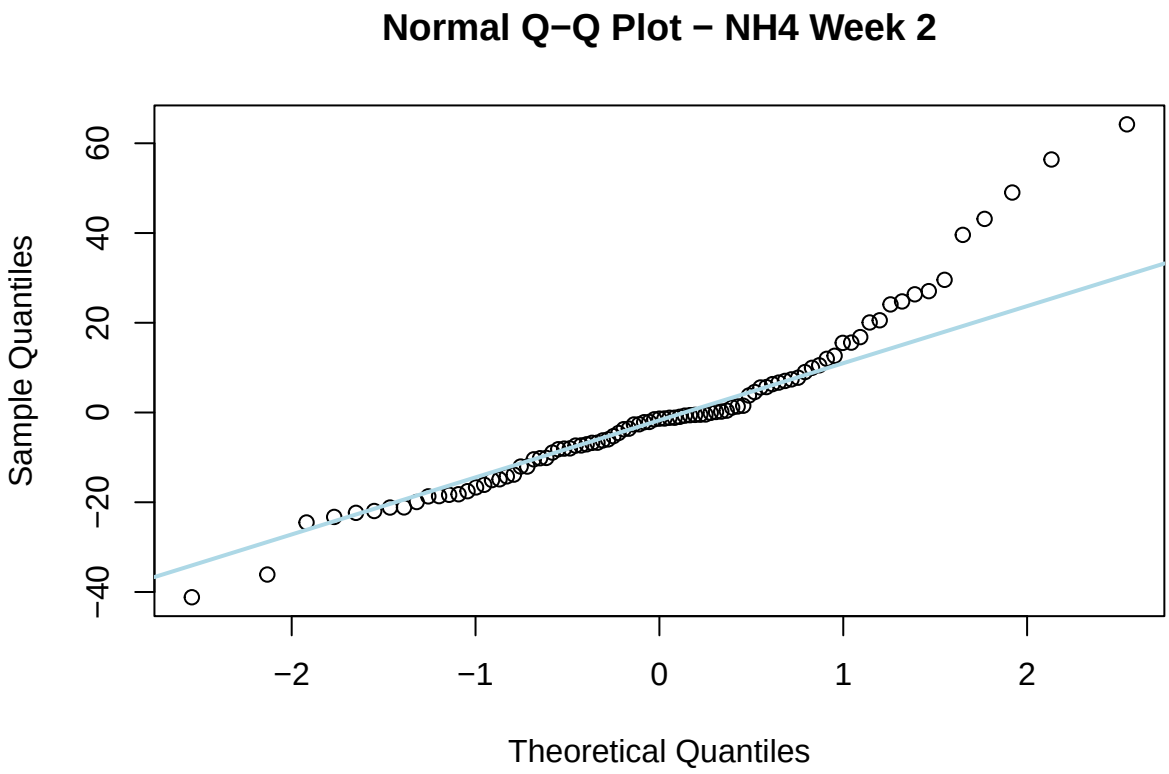


Quantile-Quantile plots

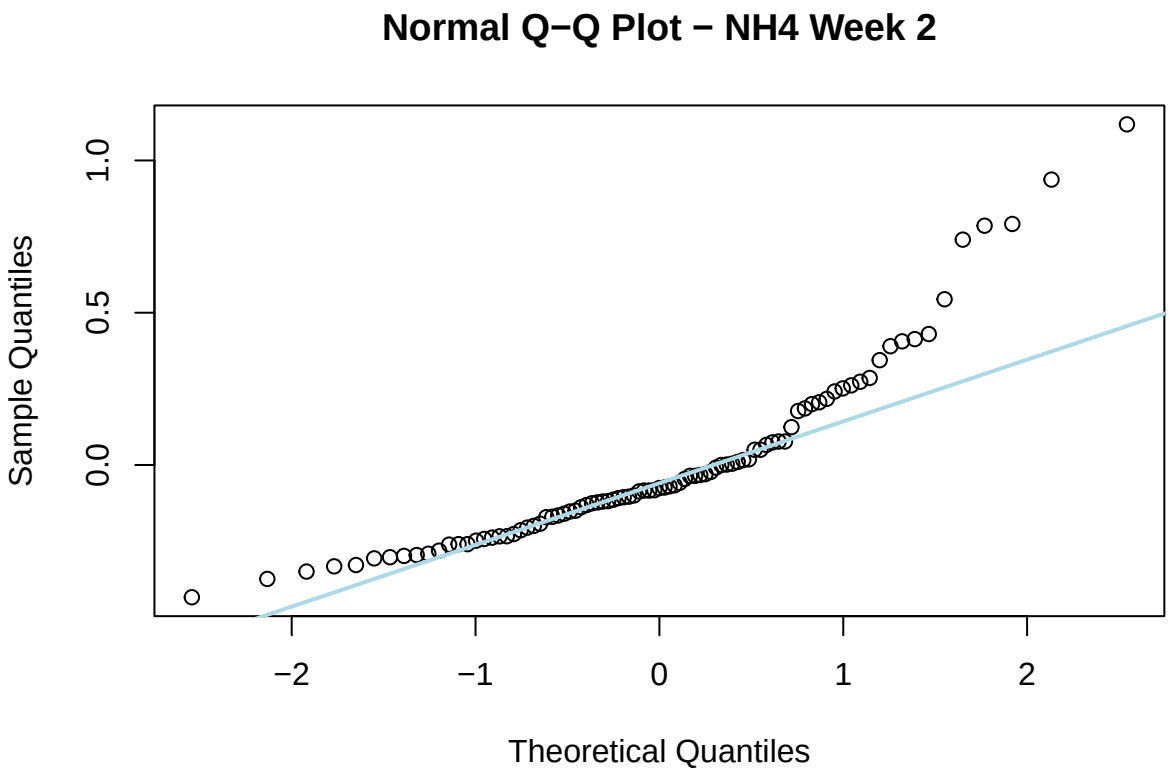
```
qqnorm(anova_NH4_wk2$residuals, main = "Normal Q-Q Plot - NH4 Week 2")
qqline(anova_NH4_wk2$residuals, col = "lightblue", lwd = 2)
```




```r
qqnorm(anova_NO3_wk2$residuals, main = "Normal Q-Q Plot - NH4 Week 2")
qqline(anova_NO3_wk2$residuals, col = "lightblue", lwd = 2)
```

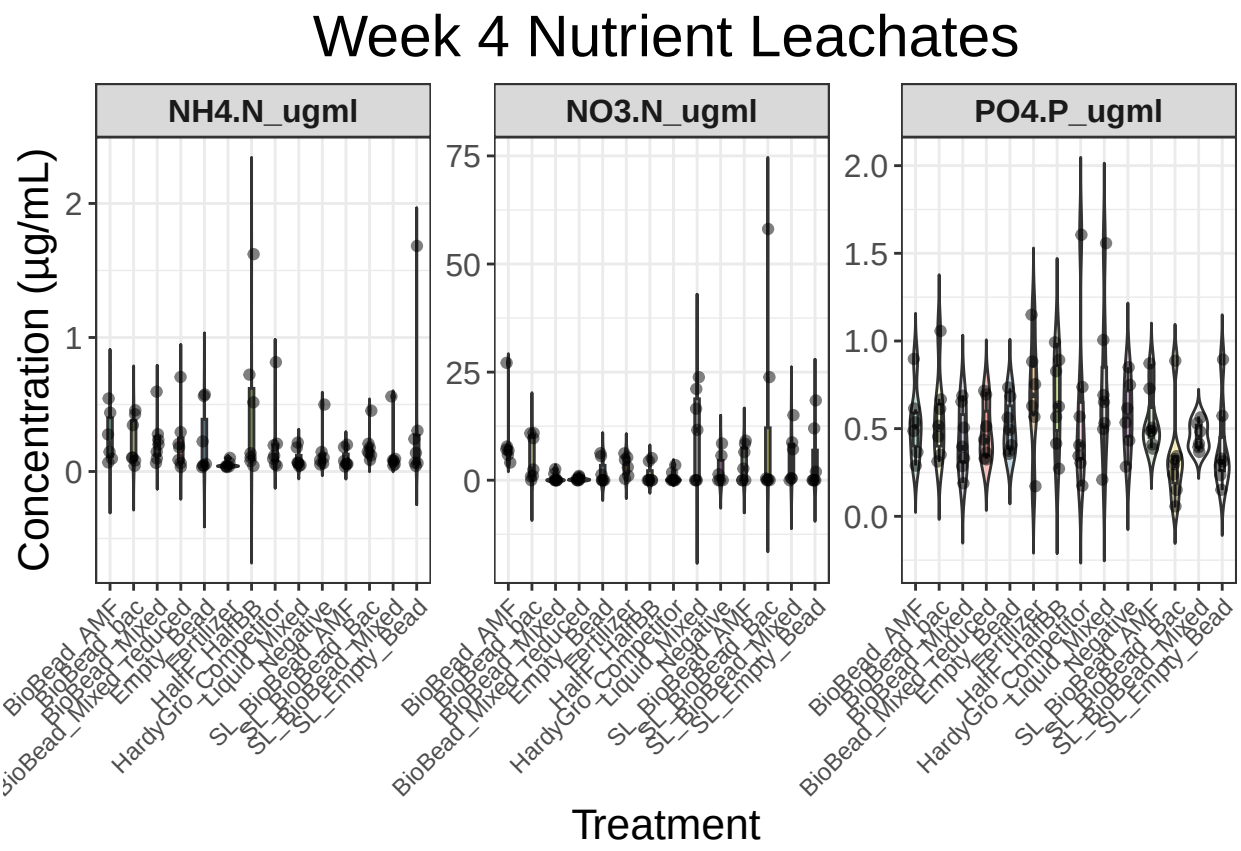



```r
qqnorm(anova_PO4_wk2$residuals, main = "Normal Q-Q Plot - NH4 Week 2")
qqline(anova_PO4_wk2$residuals, col = "lightblue", lwd = 2)
```


Normal Q−Q Plot − NH4 Week 2
Sample Quantiles
0.0
0.5
1.0
−2
−1
0
1
2
Theoretical Quantiles

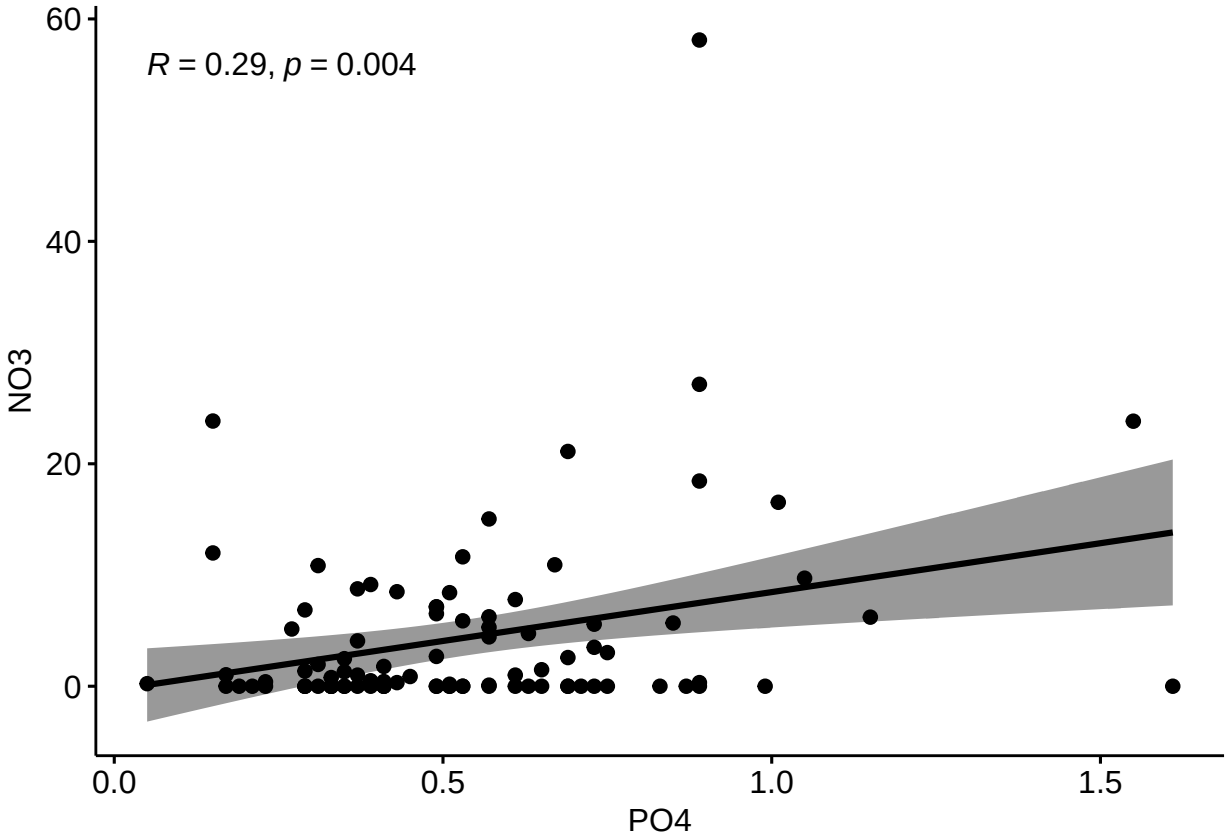
Week 4 Analysis

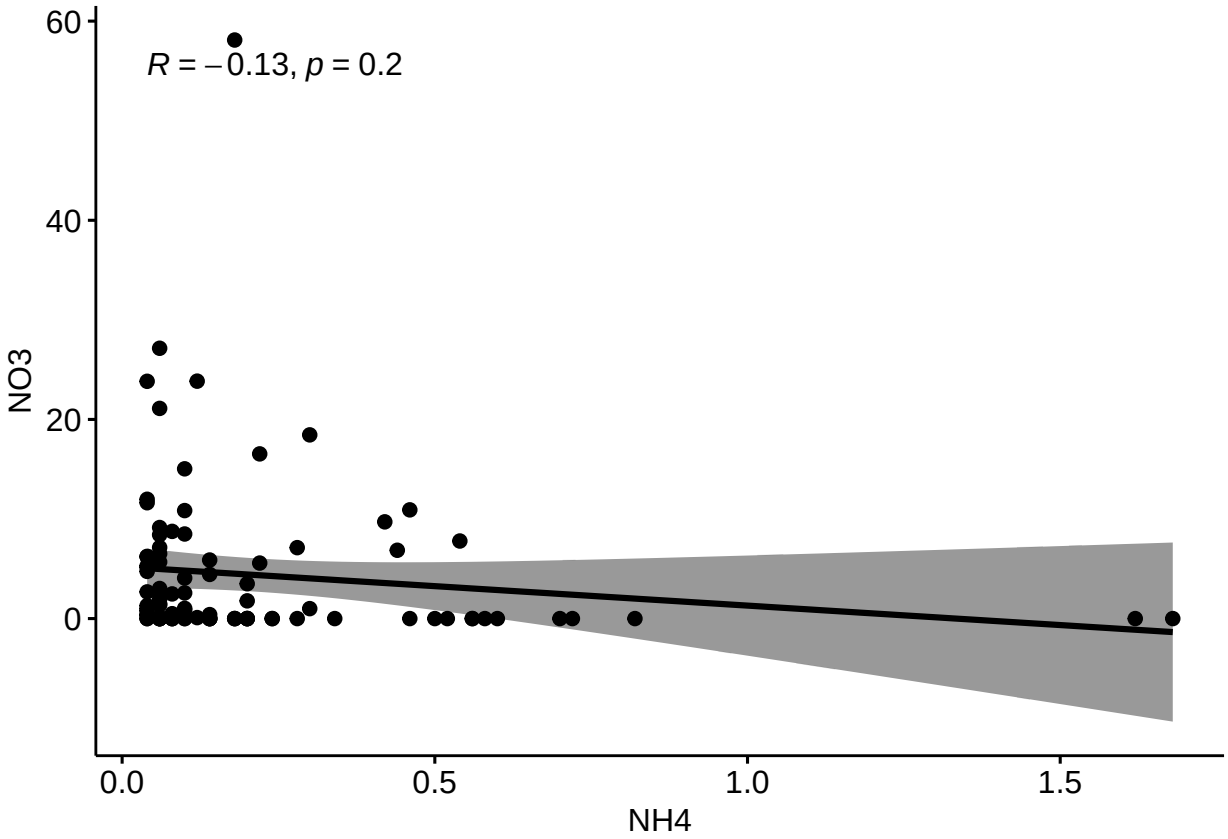
Faceted violin plot



Correlation scatterplots

Correlation scatterplots were generated to assess the correlations between nitrate and phosphate and ammonium.


R = 0.29, p = 0.004
60
40
20
0
NO3
0.0
0.5
1.0
1.5
PO4



One Way ANOVA

```
summary(anova_NH4_wk4)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  0.986 0.07583   0.968   0.49
## Residuals          80  6.269 0.07836
```

```
summary(anova_NO3_wk4)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13   1271   97.74   1.572  0.111
## Residuals          80   4973   62.16
```

```
summary(anova_PO4_wk4)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  1.133 0.08716   1.192  0.301
## Residuals          80  5.850 0.07312
```

There were no significant differences among treatment groups for ammonium, nitrate, or phosphate concentrations in leachate.

Tukey Test

```
tukey_NH4_wk4 <- TukeyHSD(x=anova_NH4_wk4, ordered=TRUE, conf.level=0.95)
#print(tukey_NH4_wk4)

tukey_NO3_wk4 <- TukeyHSD(x=anova_NO3_wk4, ordered=TRUE, conf.level=0.95)
#print(tukey_NO3_wk4)

tukey_PO4_wk4 <- TukeyHSD(x=anova_PO4_wk4, ordered=TRUE, conf.level=0.95)
#print(tukey_PO4_wk4)
```

There were no significant differences among treatment groups for ammonium, nitrate, or phosphate concentrations in leachate.

Shapiro Wilk test for normality

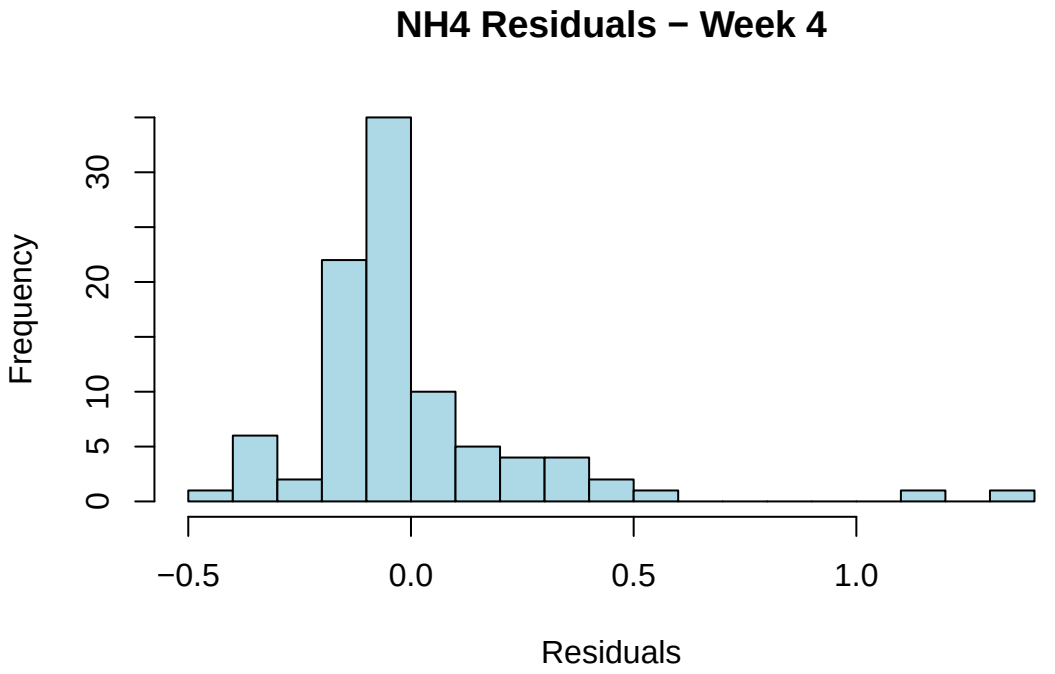
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NH4_wk4$residuals
## W = 0.77091, p-value = 8.526e-11

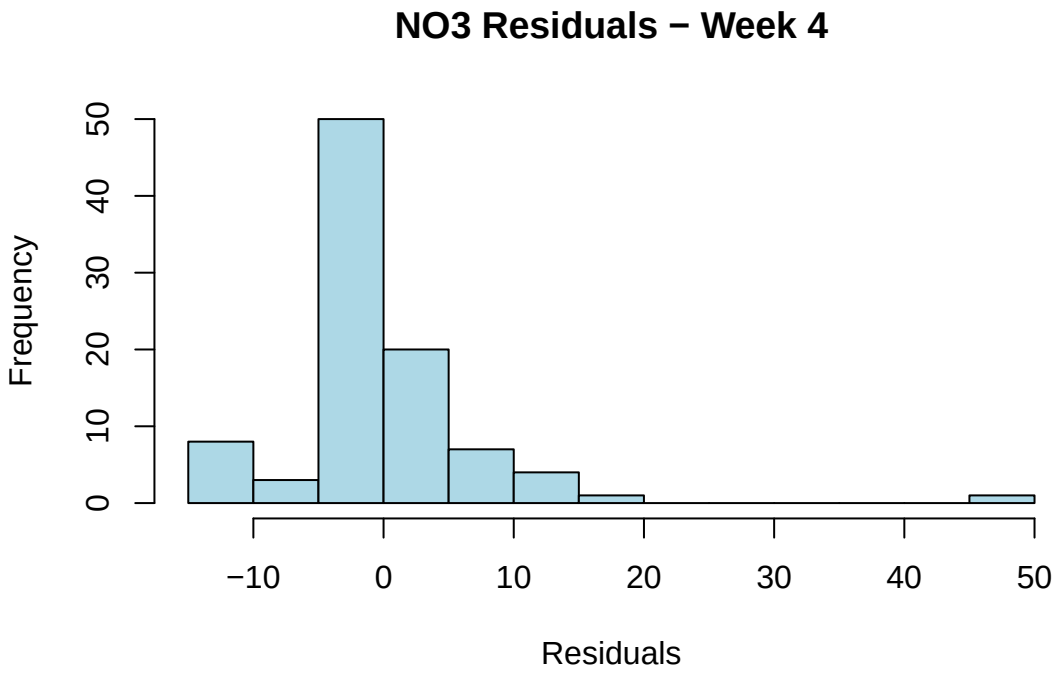
##
##  Shapiro-Wilk normality test
##
## data:  anova_NO3_wk4$residuals
## W = 0.77616, p-value = 1.19e-10

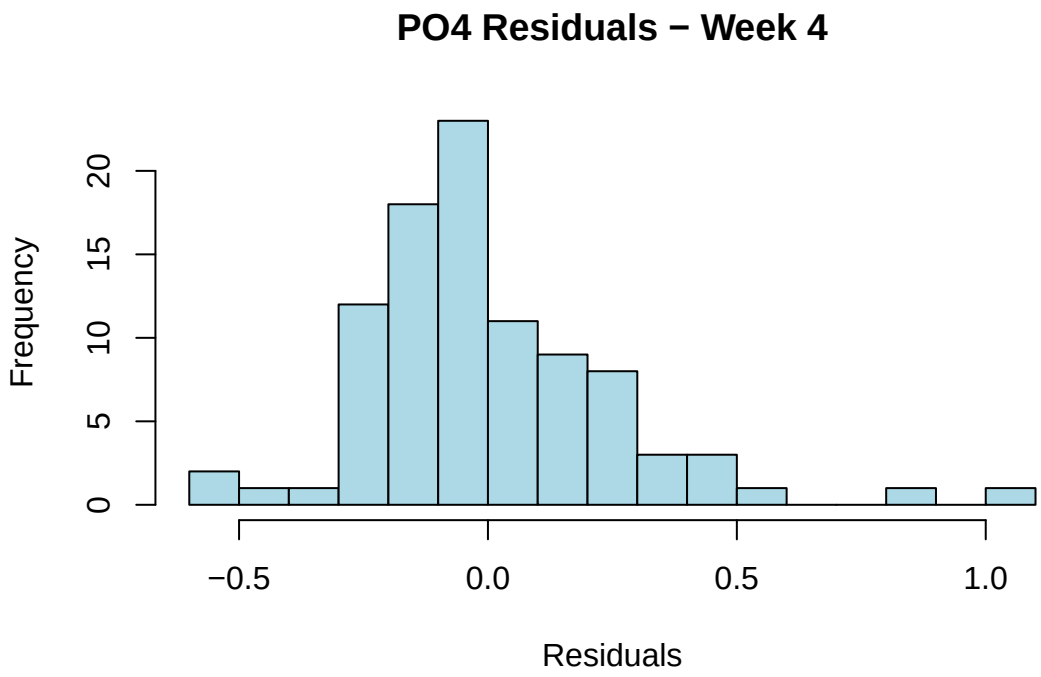
##
##  Shapiro-Wilk normality test
##
## data:  anova_PO4_wk4$residuals
## W = 0.92579, p-value = 4.938e-05
```

Similar to week 2, there are similarly high W values (0.77091, 0.77616, 0.92579) and very low p values. This indicates that the data are not normal.

Residuals distribution

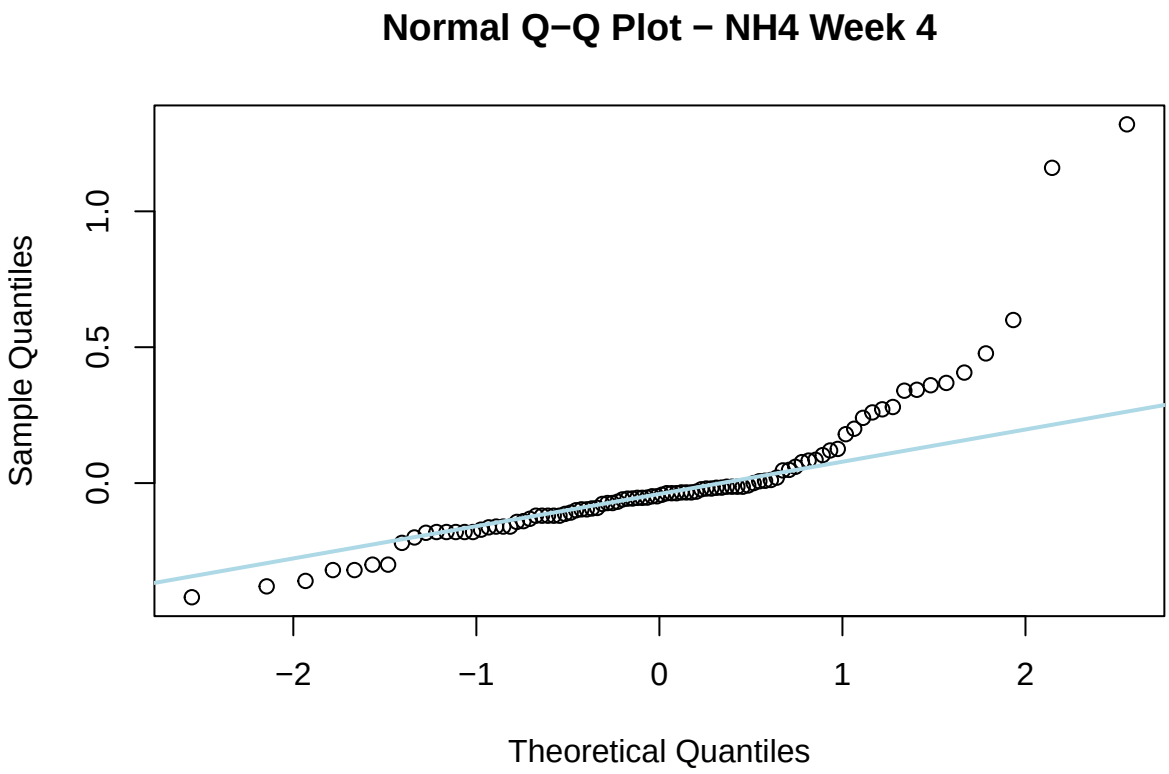




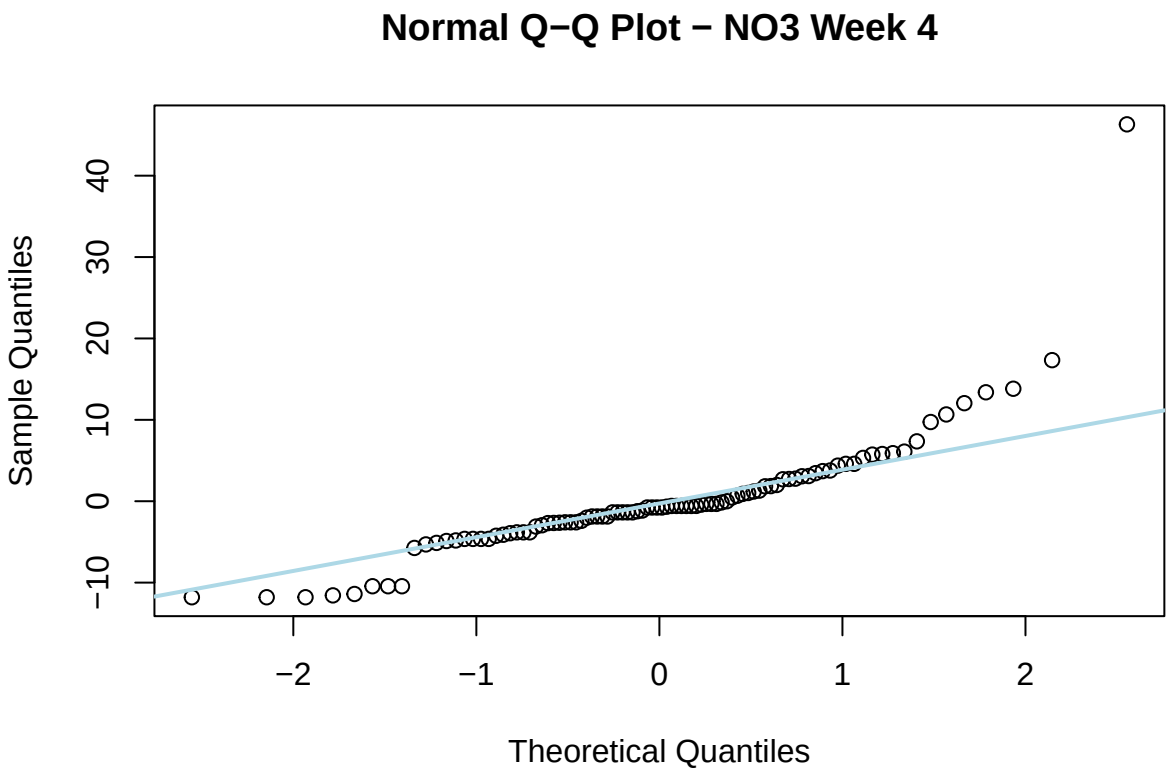


Quantile-Quantile plots

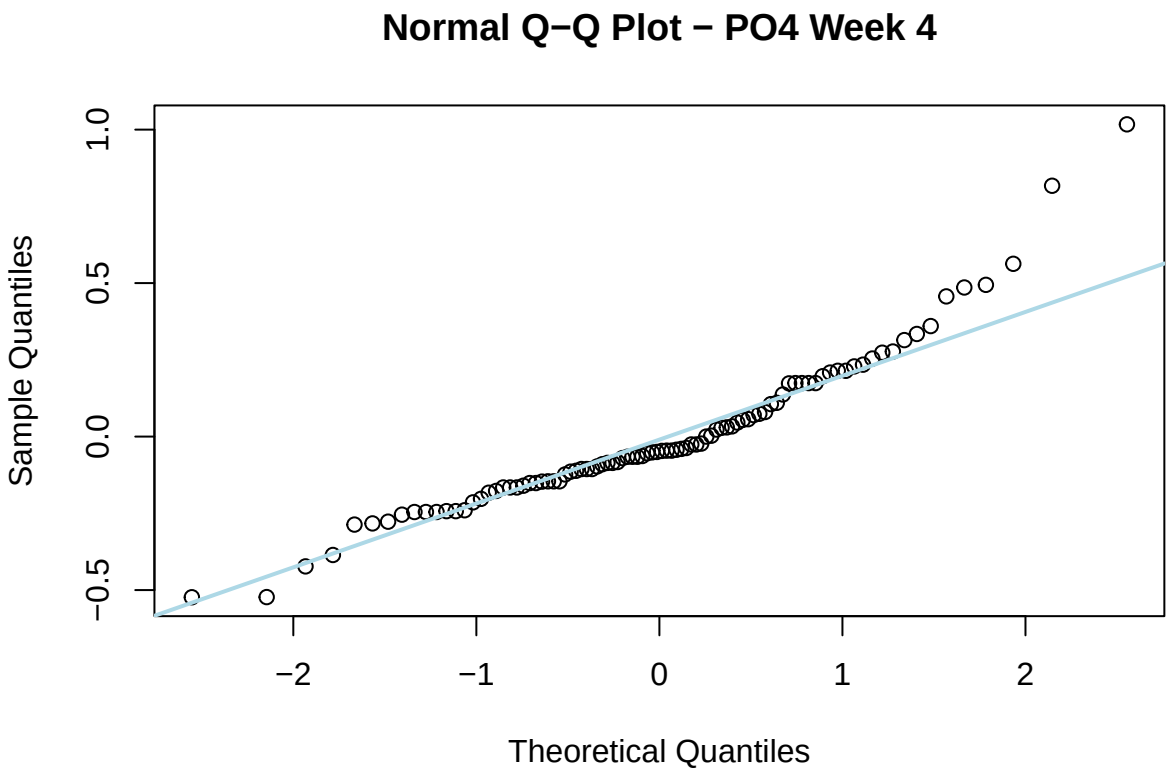
```
qqnorm(anova_NH4_wk4$residuals, main = "Normal Q-Q Plot - NH4 Week 4")
qqline(anova_NH4_wk4$residuals, col = "lightblue", lwd = 2)
```




```r
qqnorm(anova_NO3_wk4$residuals, main = "Normal Q-Q Plot - NO3 Week 4")
qqline(anova_NO3_wk4$residuals, col = "lightblue", lwd = 2)
```

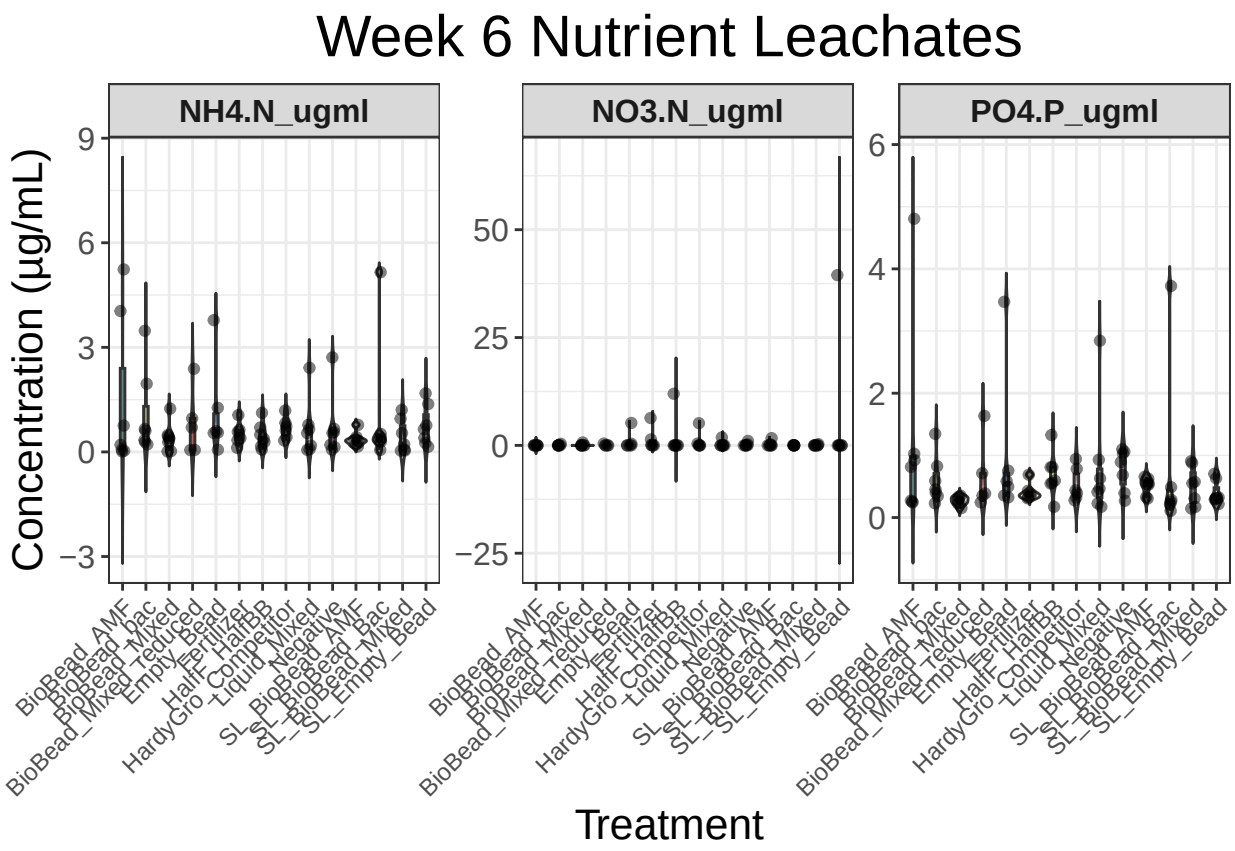



```r
qqnorm(anova_PO4_wk4$residuals, main = "Normal Q-Q Plot - PO4 Week 4")
qqline(anova_PO4_wk4$residuals, col = "lightblue", lwd = 2)
```


Normal Q−Q Plot − PO4 Week 4
Sample Quantiles
Theoretical Quantiles
1.0
0.5
0.0
−0.5
−2
−1
0
1
2

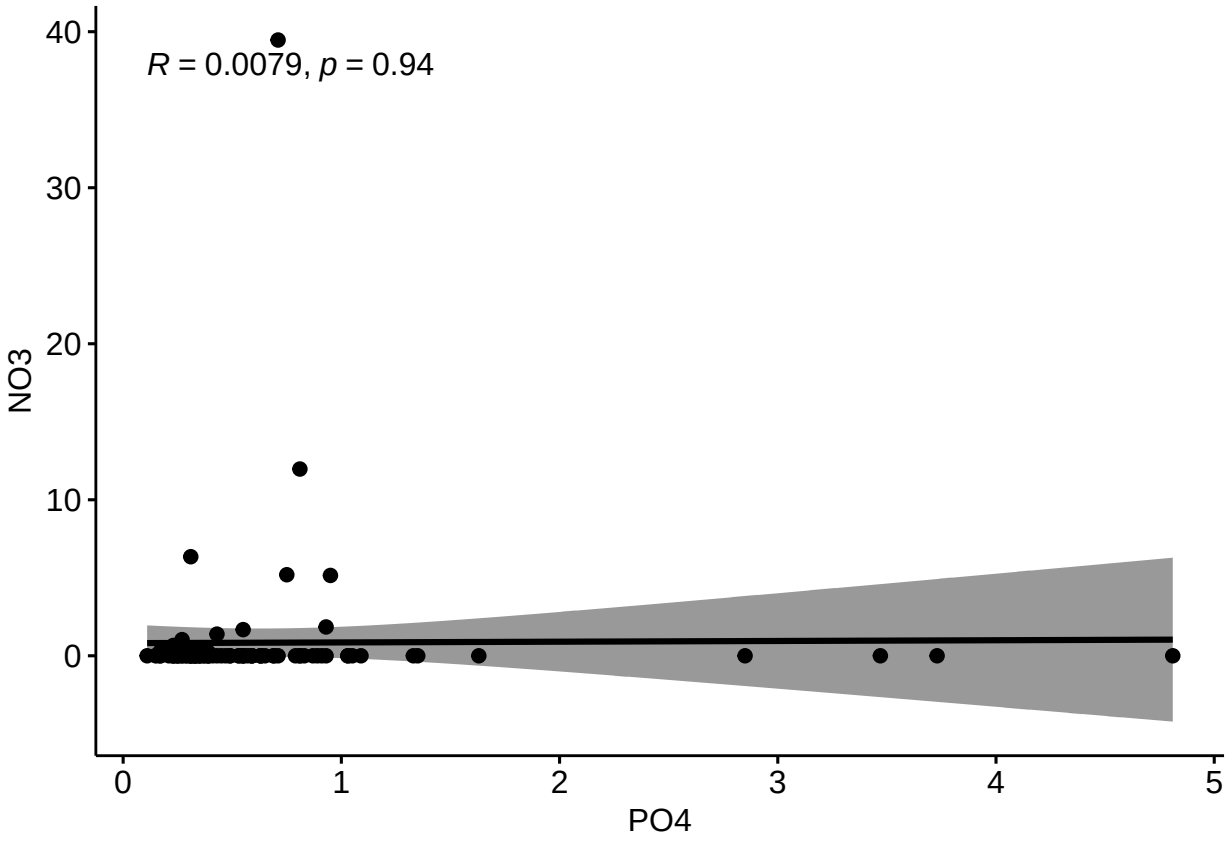
Week 6 Analysis

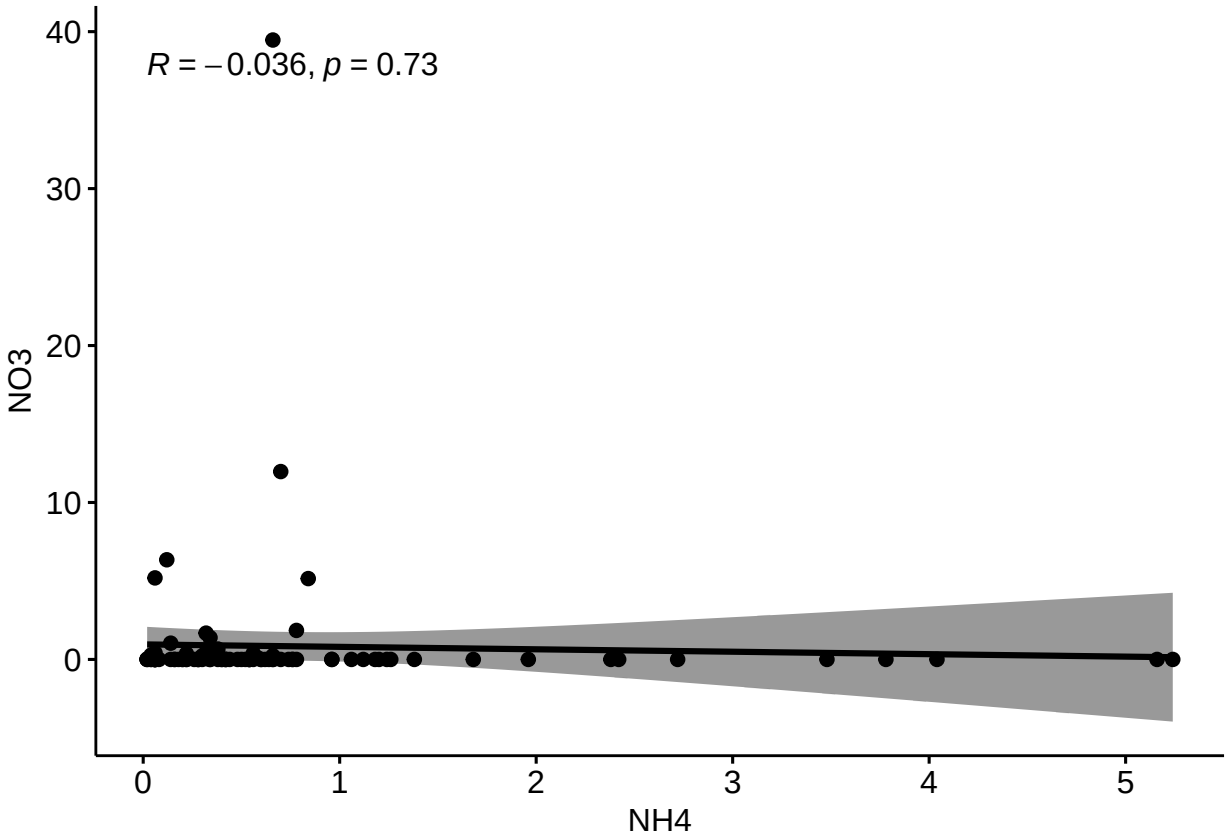
Faceted violin plot



Correlation scatterplots

Correlation scatterplots were generated to assess the correlations between nitrate and phosphate and ammonium.


R = 0.0079, p = 0.94
NO3
PO4
40
30
20
10
0
0
1
2
3
4
5



One Way ANOVA

```
summary(anova_NH4_wk6)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13   9.80   0.754   0.685  0.772
## Residuals          78  85.87   1.101
```

```
summary(anova_NO3_wk6)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  200.1   15.39    0.78  0.679
## Residuals          78 1540.3   19.75
```

```
summary(anova_PO4_wk6)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13   5.52  0.4245   0.758  0.701
## Residuals          78  43.70  0.5603
```

There were no significant differences among treatment groups for ammonium, nitrate, or phosphate concentrations in leachate.

Tukey Test

There were no significant differences among treatment groups for ammonium, nitrate, or phosphate concentrations in leachate.

Shapiro Wilk test for normality

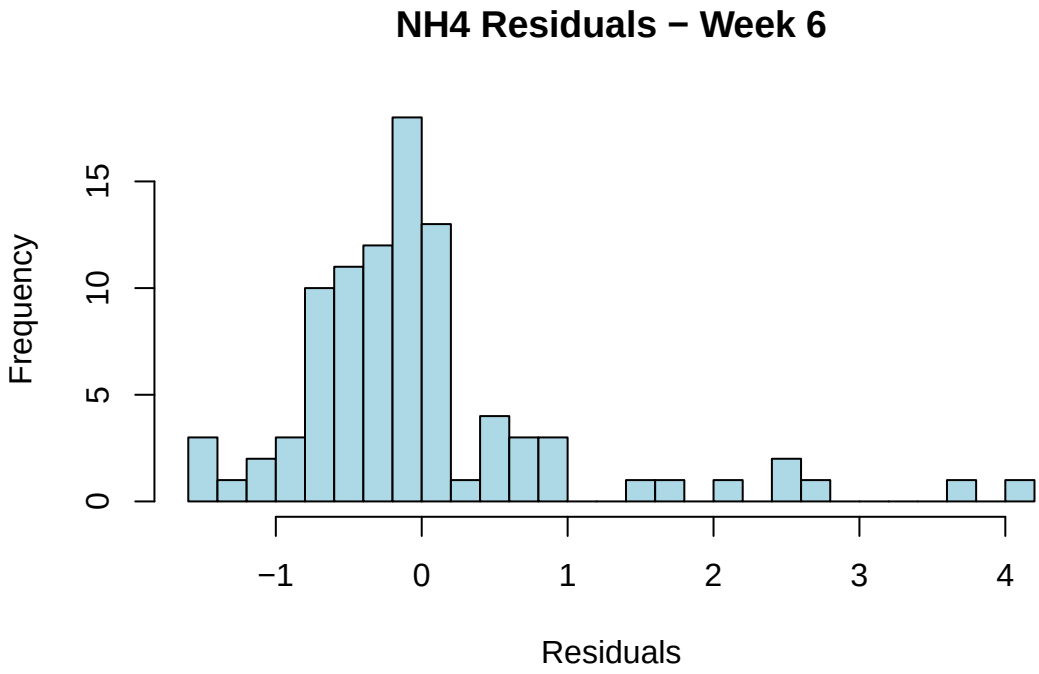
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NH4_wk6$residuals
## W = 0.80185, p-value = 8.809e-10
```

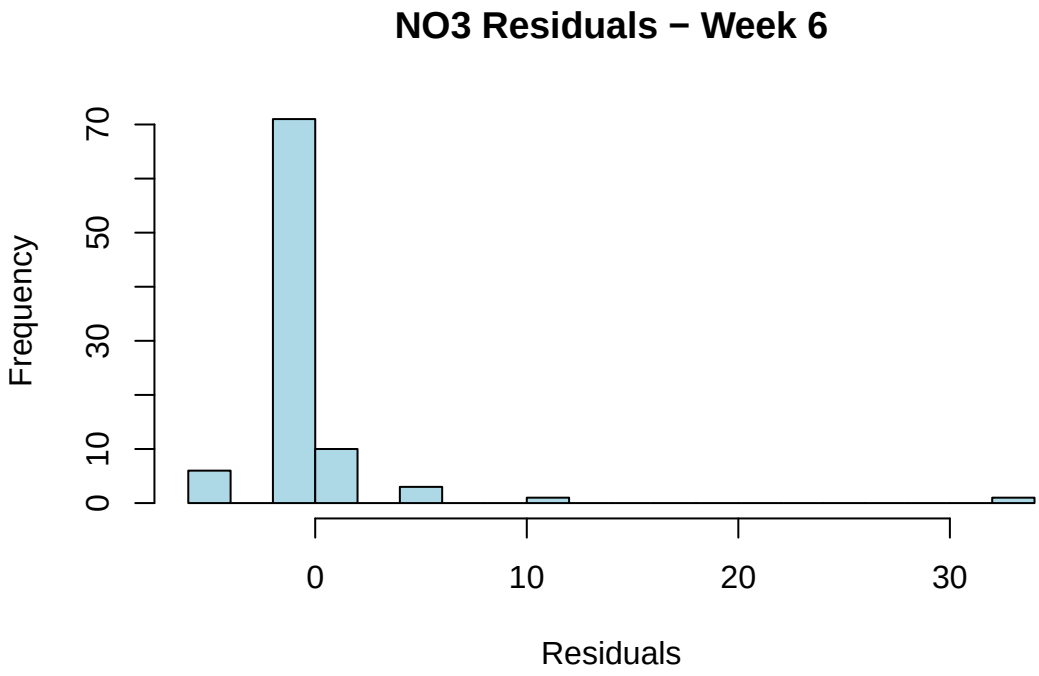
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NO3_wk6$residuals
## W = 0.3975, p-value < 2.2e-16
```

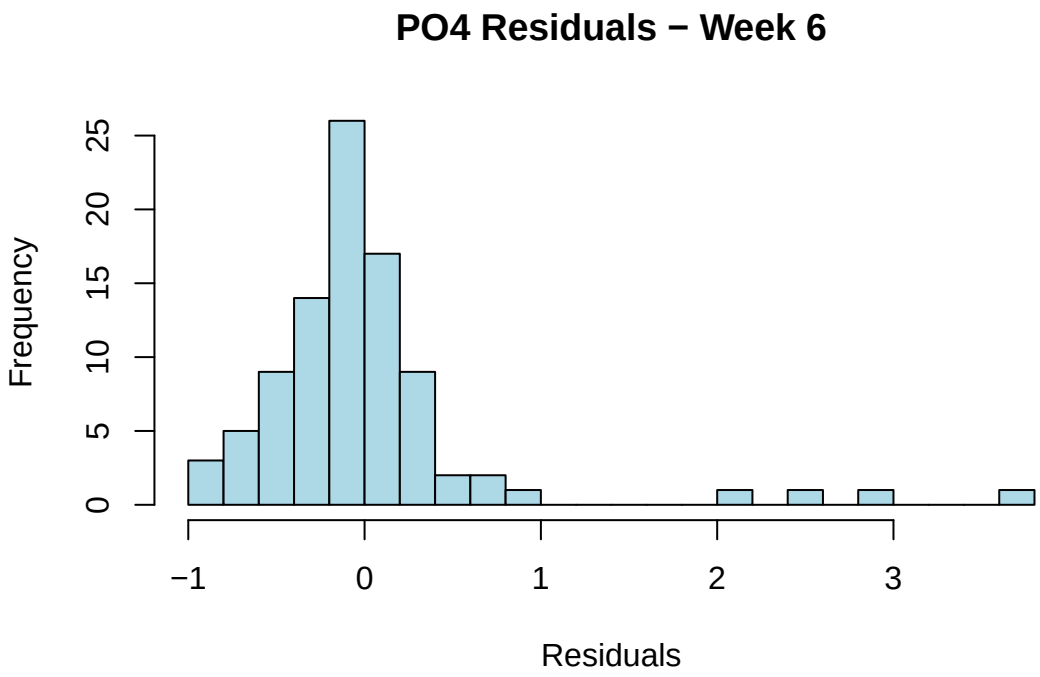
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_PO4_wk6$residuals
## W = 0.68731, p-value = 1.064e-12
```

Similar to week 2, there are similarly high W values (0.80185, 0.3975, 0.92579) and very low p values. This indicates that the data are not normal.

Residuals distribution

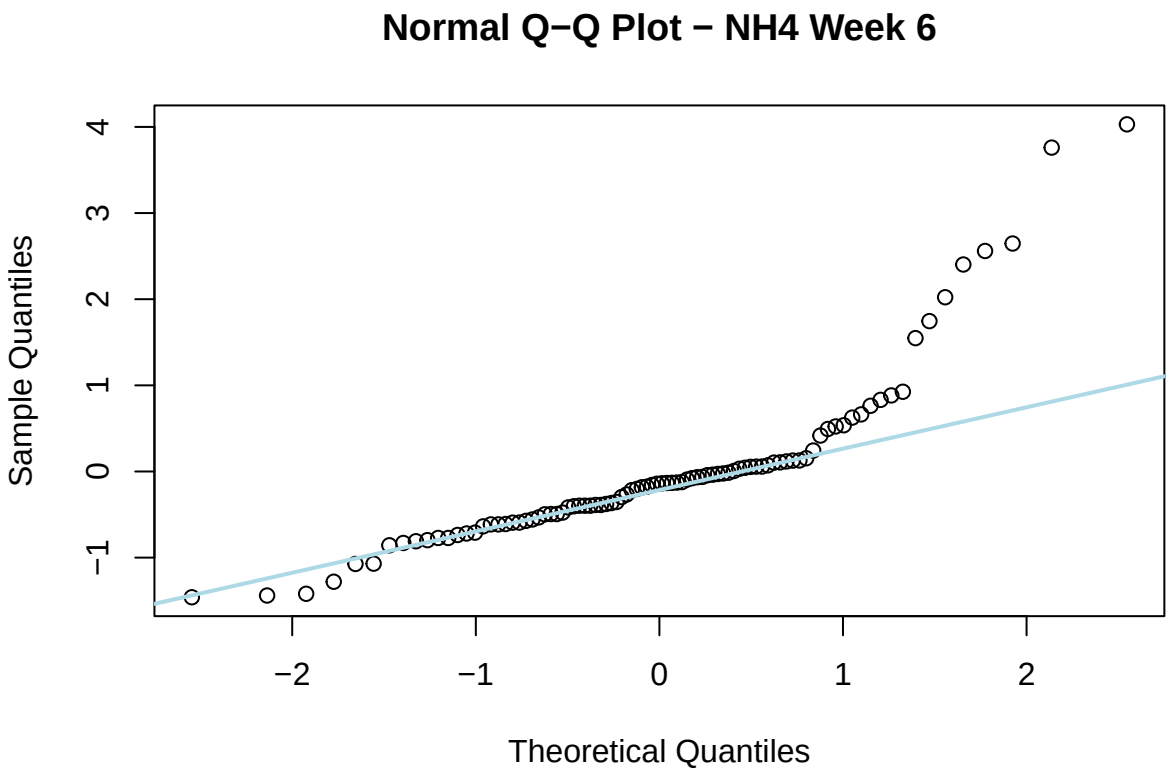




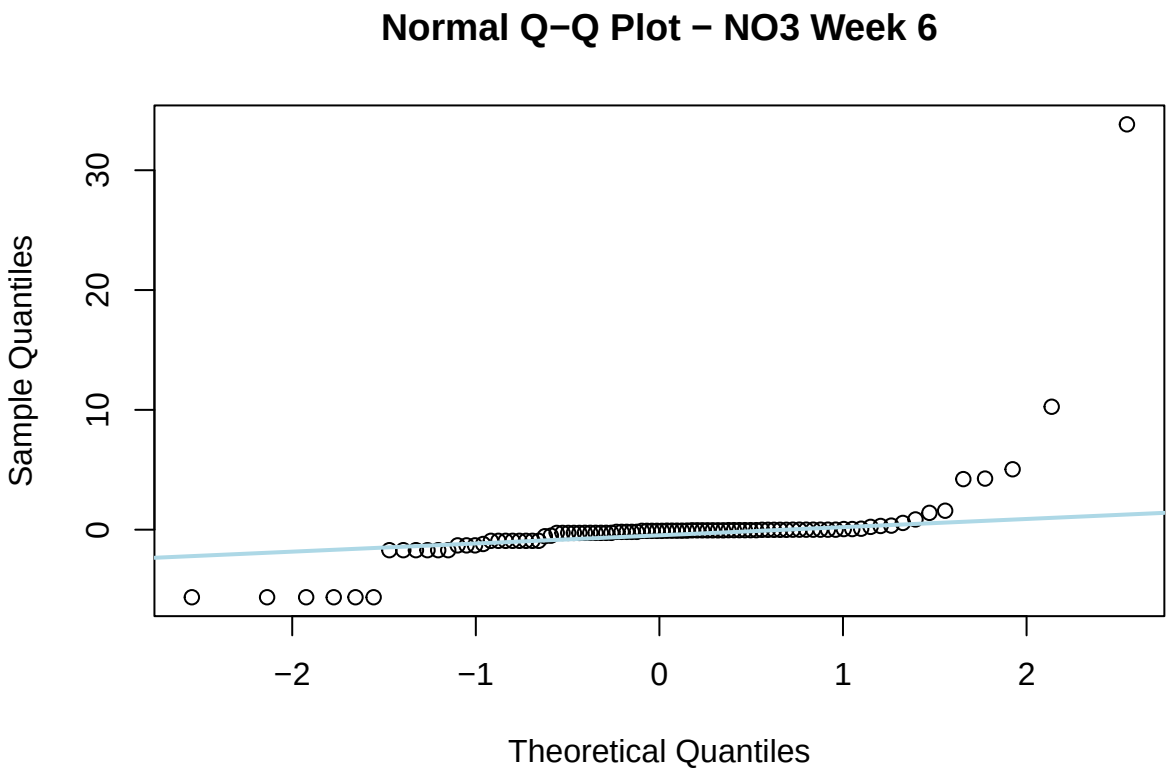


Quantile-Quantile plots

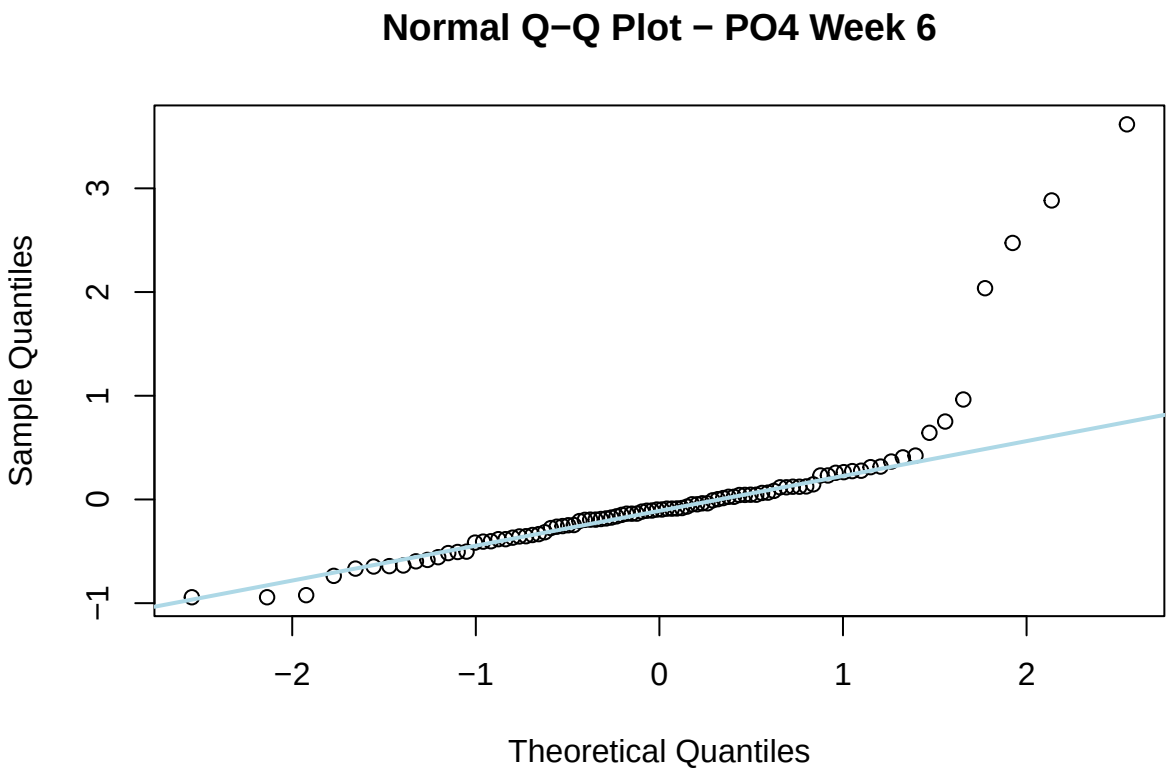
```
qqnorm(anova_NH4_wk6$residuals, main = "Normal Q-Q Plot - NH4 Week 6")
qqline(anova_NH4_wk6$residuals, col = "lightblue", lwd = 2)
```




```r
qqnorm(anova_NO3_wk6$residuals, main = "Normal Q-Q Plot - NO3 Week 6")
qqline(anova_NO3_wk6$residuals, col = "lightblue", lwd = 2)
```

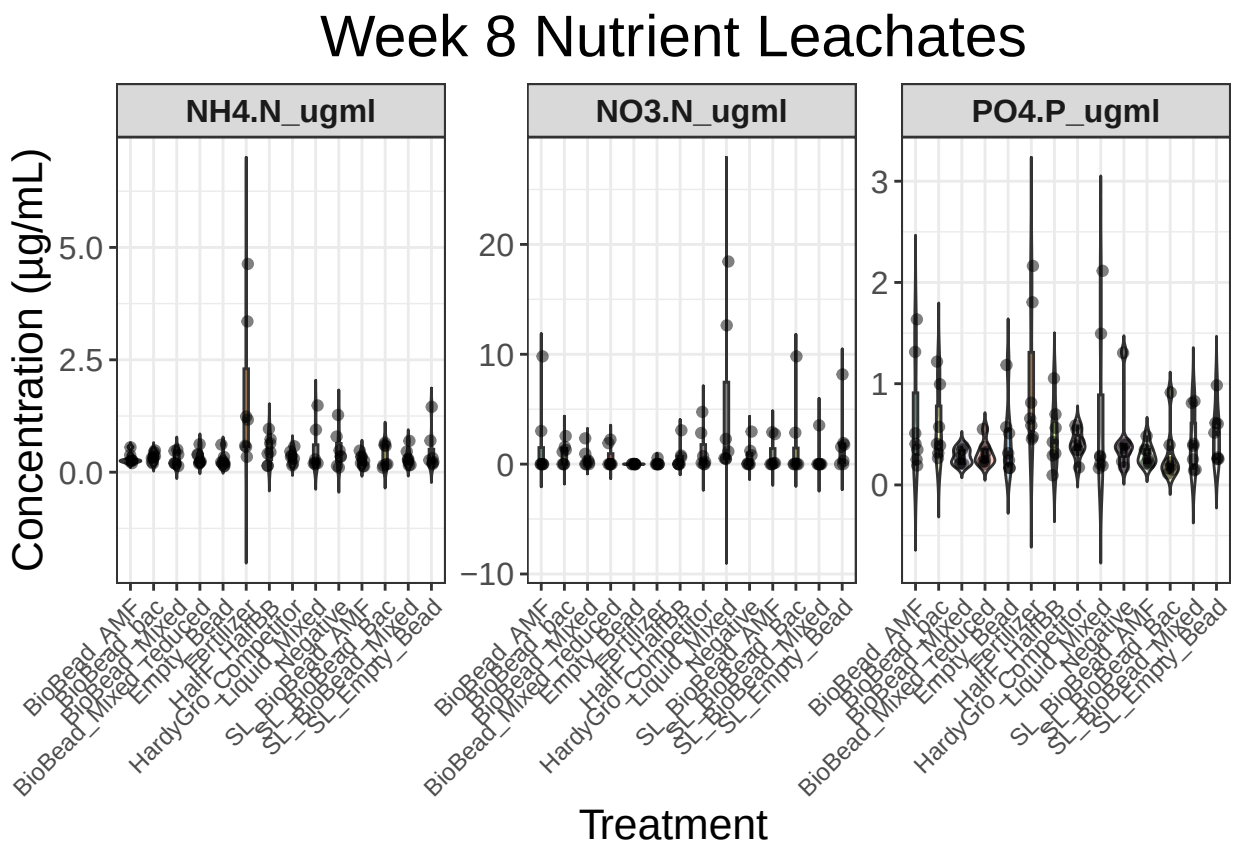



```r
qqnorm(anova_PO4_wk6$residuals, main = "Normal Q-Q Plot - PO4 Week 6")
qqline(anova_PO4_wk6$residuals, col = "lightblue", lwd = 2)
```


Normal Q−Q Plot − PO4 Week 6
Sample Quantiles
Theoretical Quantiles
−1
0
1
2
3
−2
−1
0
1
2

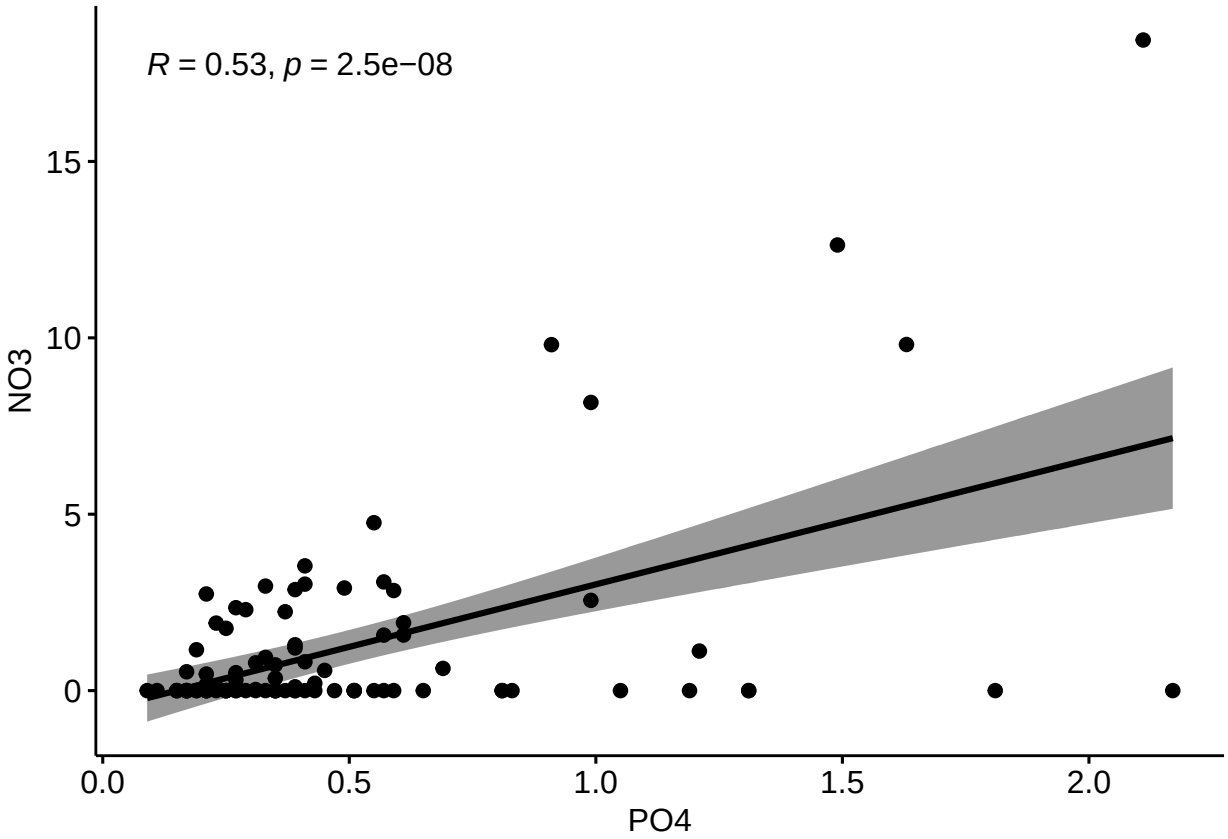
Week 8 Analysis

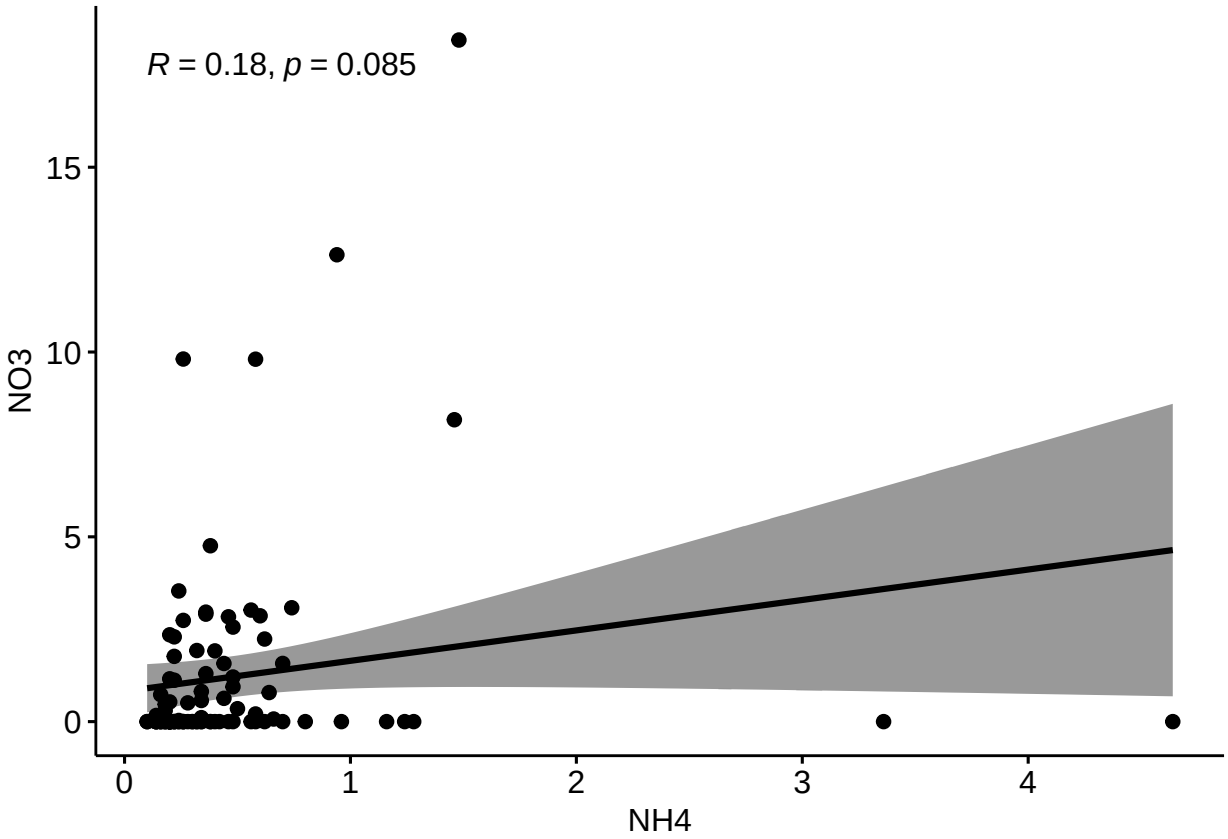
Faceted violin plot



Correlation scatterplots

Correlation scatterplots were generated to assess the correlations between nitrate and phosphate and ammonium.


R = 0.53, p = 2.5e−08
NO3
PO4
0
5
10
15
0.0
0.5
1.0
1.5
2.0



One Way ANOVA

```
summary(anova_NH4_wk8)
```

```
##                    Df Sum Sq Mean Sq F value   Pr(>F)
## Treatment_Category 13  12.11  0.9313   3.524 0.000213 ***
## Residuals          84  22.20  0.2643
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(anova_NO3_wk8)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  151.6  11.662   1.612  0.098 .
## Residuals          84  607.8   7.236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(anova_PO4_wk8)
```

```
##                    Df Sum Sq Mean Sq F value Pr(>F)
## Treatment_Category 13  3.401  0.2616   1.642 0.0899 .
```

```
## Residuals          84 13.389  0.1594
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There were no significant differences among treatment groups for nitrate or phosphate concentrations in leachate. However ammonium shows highly significant differences among treatments.

Tukey Test

There were no significant differences among treatment groups for nitrate or phosphate concentrations in leachate.

For ammonium, there were significant concentration differences between the following treatments (along with p adj values):

- Fertilizer-SL_Empty_Bead 0.0020809
- Fertilizer-SL_BioBead_AMF 0.0001264
- Fertilizer-SL_BioBead_Mixed 0.0002748
- Fertilizer-Liquid_Mixed 0.0027988
- Fertilizer-Negative 0.0027988
- Fertilizer-Empty_Bead 0.0001212

Shapiro Wilk test for normality

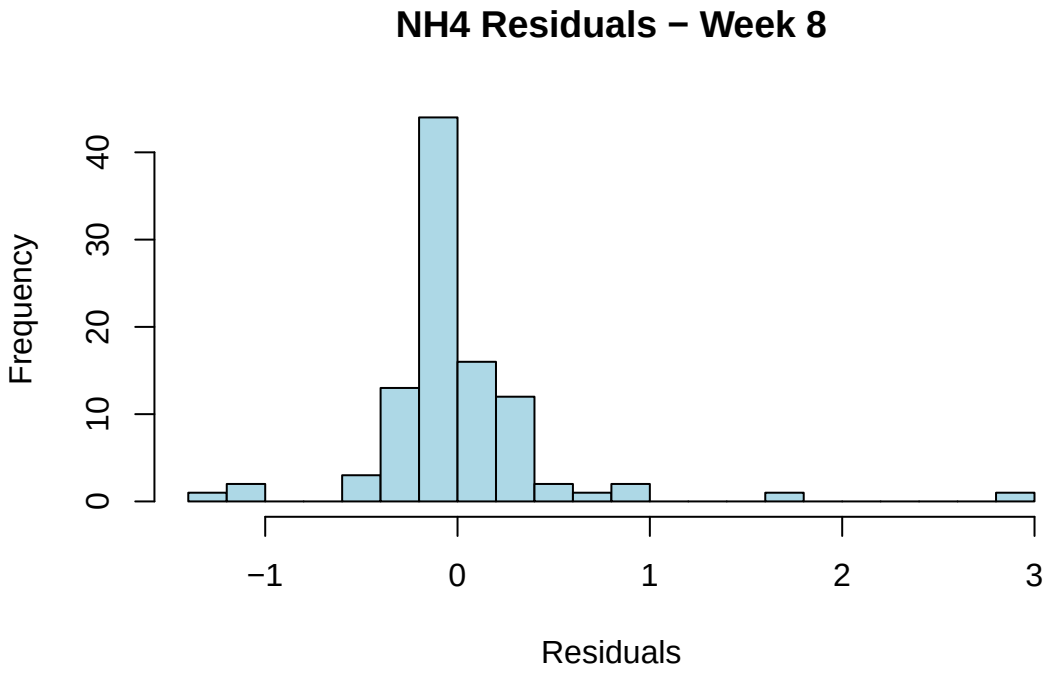
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NH4_wk8$residuals
## W = 0.72636, p-value = 3.185e-12
```

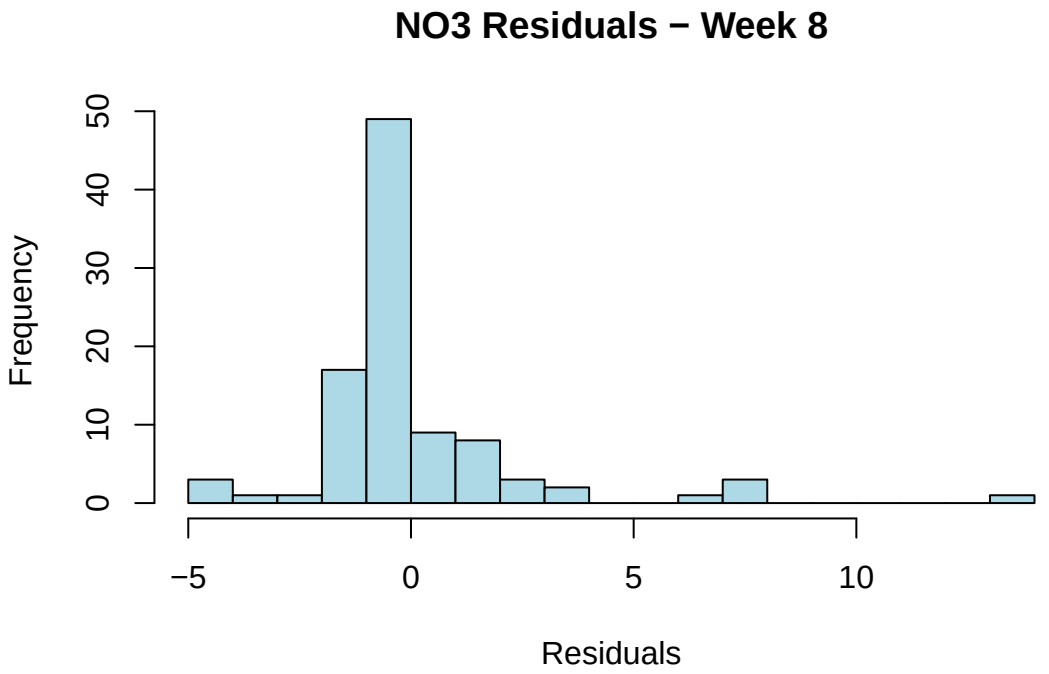
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_NO3_wk8$residuals
## W = 0.74704, p-value = 1.064e-11
```

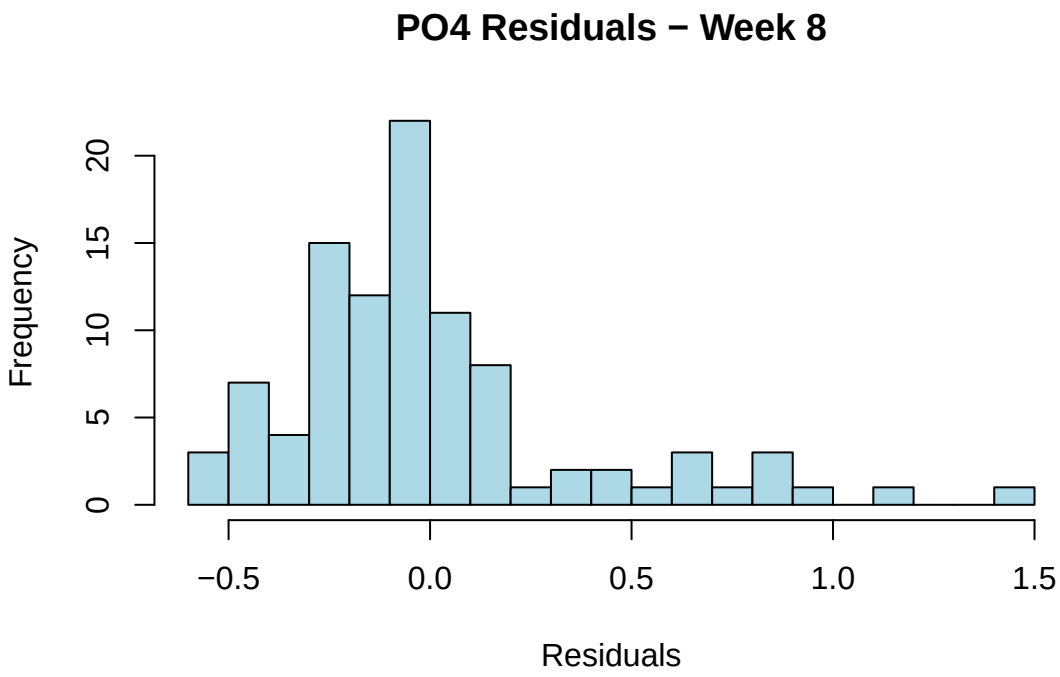
```
##
##  Shapiro-Wilk normality test
##
## data:  anova_PO4_wk8$residuals
## W = 0.87171, p-value = 1.038e-07
```

Similar to previous weeks, there are similarly very low p values. This indicates that the data are not normal.

Residuals distribution

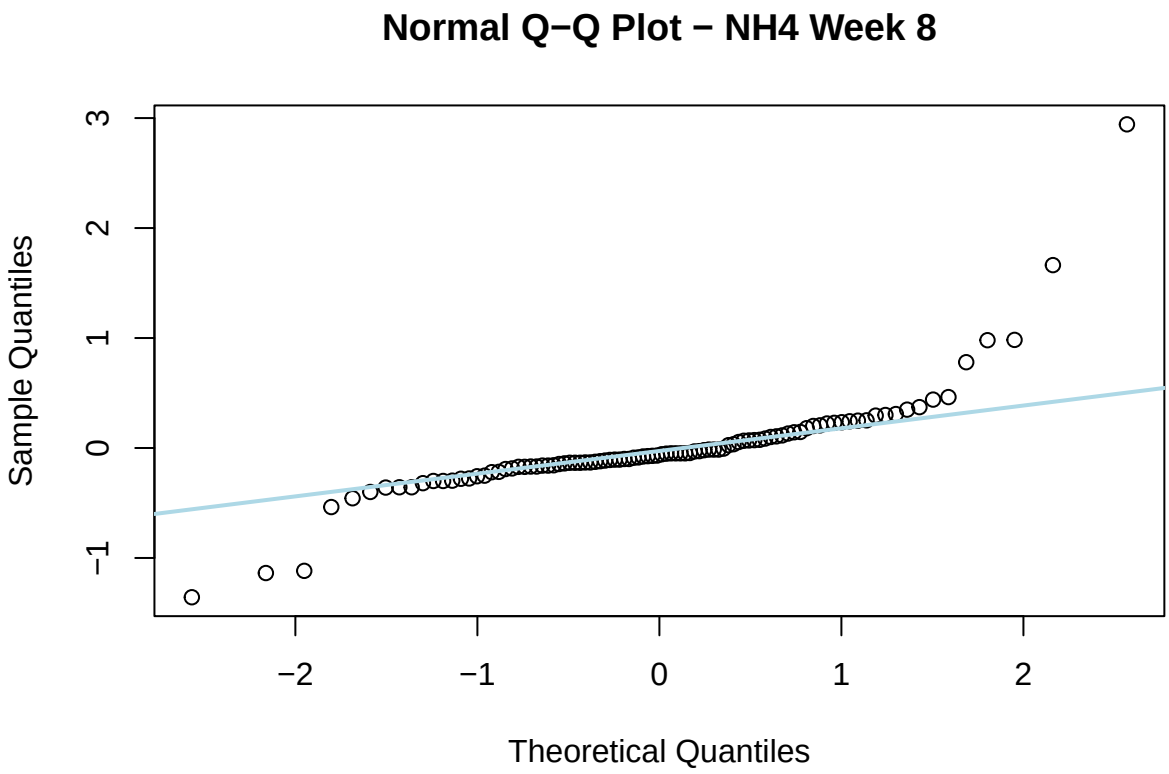




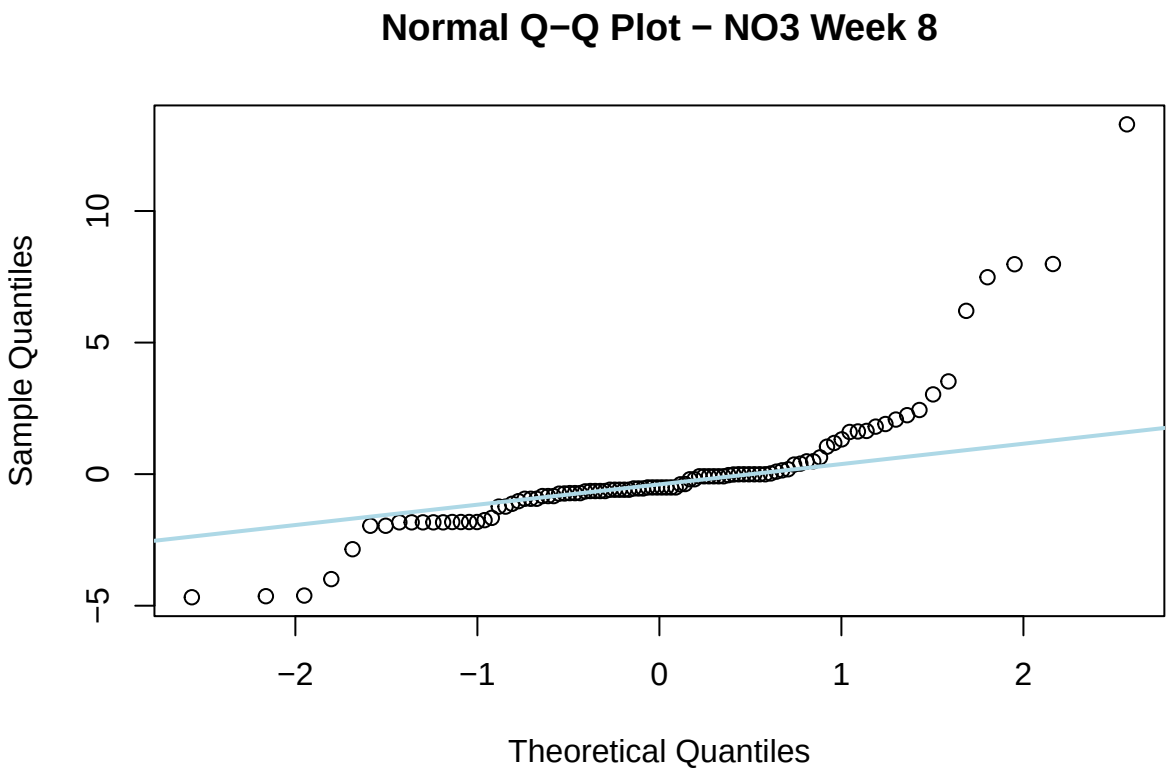


Quantile-Quantile plots

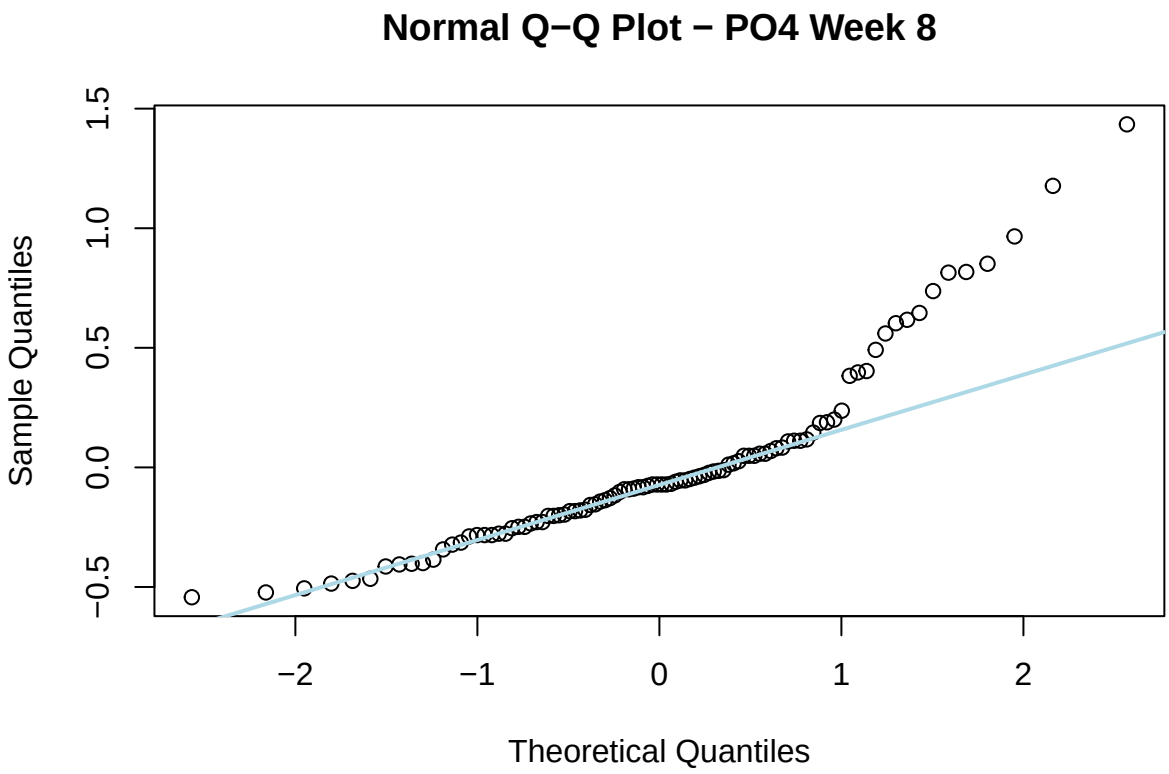
```
qqnorm(anova_NH4_wk8$residuals, main = "Normal Q-Q Plot - NH4 Week 8")
qqline(anova_NH4_wk8$residuals, col = "lightblue", lwd = 2)
```




```r
qqnorm(anova_NO3_wk8$residuals, main = "Normal Q-Q Plot - NO3 Week 8")
qqline(anova_NO3_wk8$residuals, col = "lightblue", lwd = 2)
```




```r
qqnorm(anova_PO4_wk8$residuals, main = "Normal Q-Q Plot - PO4 Week 8")
qqline(anova_PO4_wk8$residuals, col = "lightblue", lwd = 2)
```



Discussion

Leachate data is not normally distributed, hence ANOVA is not the appropriate statistical approach. Regardless, we start to see signal at week 8 with ammonium concentrations. These are mostly comparisons to synthetic fertilizer, which is expected.

Next steps

- Use alternative to ANOVA, like Non-Parametric Tests (Kruskal-Wallis) or Linear Mixed Effects Model
- Consider data transformation (e.g. relativizing to max) to address normality issues
- Consider multivariate approaches to connect with other data (soil moisture, greenhouse gases, soil ions)